Nyra MES Core — database structure and stored data

Snapshot 2026-10-05 17:06 · database postgres on nyra-mes-core · tenant GR_CARBONS (GR Carbons) · transactional rows are GR Carbons only. Passwords and secrets are redacted. Large tables are capped.

1. How data is stored

Nyra MES Core is one PostgreSQL database with schemas: **core** (tenancy), **template** (industry packs), **mst** (masters), **trn** (orders / lots / dispatch), **exec** (shop-floor runs), **qms** (quality), **kg** (triplet ontology). Activated Carbon shop-floor capture (gate receipt, lab, RMS bags, crusher, outward, kiln bags, destoning, post-activation destinations) is kept as **JSONB metadata** on inward orders, lots, and quality rows — not as extra tables. Triplet subjects stay **urn:nyra:ac:inward:{ORD-*}** and **urn:nyra:ac:lot:{LOT-*}**.

Workspaces: GR_CARBONS_ACTIVATED_CARBON (ACTIVE).

2. Table catalogue (GR_CARBONS counts)

Table	What is stored	Rows (scope)	Shown
core.app_user	Login identities (Cognito email / sub). No passwords stored here.	0	0
core.capability	Platform capability catalog.	10	10
core.project_workspace	Plant workspace. GR Carbons uses GR_CARBONS_ACTIVATED_CARBON.	1	1
core.tenant	Plant / customer tenancy. GR Carbons is tenant_code GR_CARBONS.	1	1
core.user_project_access	User ↔ workspace role.	0	0
core.user_tenant_access	User ↔ tenant role (MES_RMS, MES_Activation, MES_Admin, …).	0	0
template.industry_capability	Which platform capabilities each industry template enables.	10	10
template.industry_template	Industry packs (Activated Carbon, Automobile OEM, …).	2	2
template.raw_material_parameter_profile	Shared AC raw-material QC / yield bands (not per-route).	21	21
mst.customer	End customers. Also used as the dummy buyer on inward (supplier receipt) POs.	1	1
mst.equipment	Physical equipment linked to a resource.	3	3
mst.modbus_register_map	See columns below.	7	7
mst.operation_resource_requirement	Which resource an operation needs.	11	11
mst.parameter_business_mapping	See columns below.	7	7
mst.product	Finished / intermediate grades (FG-AC-*).	3	3
mst.resource	Machines, rotaries, kilns, labor, material resources.	11	11
mst.routing	Process route for a product (AC standard process).	2	2
mst.routing_operation	Ordered operations on a route (kiln feed, destoning, screening, …).	16	16
mst.routing_operation_parameter	Per-operation capture parameters and limits.	0	0
mst.sensor_device	See columns below.	1	1
mst.sensor_parameter	See columns below.	7	7
mst.supplier	Raw-material vendors for gate receipt.	1	1
trn.customer_order	Inward receipts (ORD-AC-*) and customer sales POs (CUST_PO_*). Shop-floor RM state lives in metadata JSON.	24	24
trn.customer_order_line	Order lines (product + qty) for inward and sales.	34	34
trn.dispatch	Dispatch documents against a customer sales PO.	0	0
trn.feasibility_check	Feasibility header (UI gate is skipped; ontology rows may still exist).	211	80
trn.feasibility_check_detail	Per-check feasibility lines.	0	0
trn.production_lot	Lot created for ontology (LOT-AC-*). Activation / destoning / post-activation state is in metadata.	23	23

Table	What is stored	Rows (scope)	Shown
trn.production_order	Internal PO created for ontology when inward is received (not shown on shop floor).	10	10
trn.sensor_reading	See columns below.	753	80
trn.sensor_reading_value	See columns below.	5271	80
exec.operation_execution	Shop-floor operation run for a lot + routing step.	185	80
exec.parameter_reading	Numeric / text readings captured on an execution.	0	0
qms.quality_check	QC header for a lot (lab, destoning bag QC, kiln output QC, dispatch QC).	1	1
qms.quality_result	QC characteristics; bag lists and destinations often sit in metadata JSON.	0	0
kg.triple_store_staging	Triplet ontology: subject / predicate / object for inward and lot URIs.	1163	60

3. Structure and data by table

core.app_user

Login identities (Cognito email / sub). No passwords stored here.

Primary key: user_id.

Column	Type	Null	Default
user_id	UUID	NO	gen_random_uuid()
cognito_sub	VARCHAR(128)	YES	
email	VARCHAR(320)	NO	
display_name	VARCHAR(256)	YES	
is_active	BOOLEAN	NO	true
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

No rows in scope (0 total in this filter).

core.capability

Platform capability catalog.

Primary key: capability_id.

Column	Type	Null	Default
capability_id	UUID	NO	gen_random_uuid()
capability_code	VARCHAR(128)	NO	
capability_name	VARCHAR(256)	NO	
description	TEXT	YES	
category	VARCHAR(64)	YES	
is_active	BOOLEAN	NO	true
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

Data (10 of 10 row(s) in GR_CARBONS / catalog scope):

capability_code	created_at	metadata	capability_id	capability_name	description	category	is_active	updated_at
AC_RAW_MATERIAL_INWARD	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Raw Material Inward	Log and track inbound raw material receipts.	INBOUND	True	2026-05-31 03:57
AC_RAW_MATERIAL_QC	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Raw Material QC	Quality inspection of incoming raw materials.	QUALITY	True	2026-05-31 03:57
AC_KILN_FEED	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Kiln Feed	Charge kiln feed batches from prepared raw material.	PRODUCTION	True	2026-05-31 03:57
AC_KILN_ACTIVATION	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Kiln Activation	Track pyrolysis and steam activation cycles.	PRODUCTION	True	2026-05-31 03:57
AC_HOURLY_KILN_PARAMETERS	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Hourly Kiln Parameters	Capture hourly temperature, pressure, and flow readings.	EXECUTION	True	2026-05-31 03:57

capability_code	created_at	metadata	capability_id	capability_name	description	category	is_active	updated_at
AC_SCREENING	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Screening	Size classification and screening operations.	PRODUCTION	True	2026-05-31 03:57
AC_QUALITY_TESTING	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Quality Testing	Iodine number, moisture, ash, and batch quality tests.	QUALITY	True	2026-05-31 03:57
AC_YIELD_ANALYSIS	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Yield Analysis	Mass-balance and yield analytics across activation batches.	ANALYTICS	True	2026-05-31 03:57
AC_TRACEABILITY	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Traceability	Batch genealogy from raw material to finished carbon.	TRACEABILITY	True	2026-05-31 03:57
AC_KNOWLEDGE_GRAPH	2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f8000000	Knowledge Graph	Ontology and knowledge-graph views for activated carbon workflows.	INTELLIGENCE	True	2026-05-31 03:57

core.project_workspace

Plant workspace. GR Carbons uses GR_CARBONS_ACTIVATED_CARBON.

Primary key: workspace_id. Foreign keys: tenant_id → core.tenant(tenant_id).

Column	Type	Null	Default
workspace_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
workspace_code	VARCHAR(64)	NO	
workspace_name	VARCHAR(256)	NO	
status	VARCHAR(32)	NO	'ACTIVE'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

Data (1 of 1 row(s) in GR_CARBONS / catalog scope):

workspace_code	status	created_at	metadata	workspace_id	workspace_name	updated_at
GR_CARBONS_ACTIVATED_CARBON	ACTIVE	2026-05-31 13:32	{"module": "activated-carbon", "industry": "ACTIVATED_CARBON", "kiln_parameter_definitions": [{"key": "zone1_temp_c", "unit": "\u00b0C", "label": "Zone 1 Temp (\u00b0C)"}, {"key": "zone2_temp_c", "unit": "\u00b0C", "lab…	e7000000	GR Carbons — Activated Carbon	2026-10-03 12:13

core.tenant

Plant / customer tenancy. GR Carbons is tenant_code GR_CARBONS.

Primary key: tenant_id. Foreign keys: industry_template_id → template.industry_template(industry_template_id).

Column	Type	Null	Default
tenant_id	UUID	NO	gen_random_uuid()
tenant_code	VARCHAR(64)	NO	
tenant_name	VARCHAR(256)	NO	
industry_template_id	UUID	NO	

Column	Type	Null	Default
status	VARCHAR(32)	NO	'ACTIVE'::character varying
timezone	VARCHAR(64)	NO	'UTC'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

Data (1 of 1 row(s) in GR_CARBONS / catalog scope):

tenant_code	status	created_at	metadata	tenant_name	industry_template_id	timezone	updated_at
GR_CARBONS	ACTIVE	2026-05-31 13:32	{"industry": "ACTIVATED_CARBON", "production": true, "ac_inbound_admin": {"rotaries": [{"code": "ROTARY_A", "name": "Rotary A"}, {"code": "ROTARY_B", "name": "Rotary B"}], "outward_splits": [{"key": "minus20_1st_rotary_…	GR Carbons	a1000000	Asia/Kolkata	2026-10-03 12:13

core.user_project_access

User ↔ workspace role.

Primary key: access_id. Foreign keys: user_id → core.app_user(user_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
access_id	UUID	NO	gen_random_uuid()
user_id	UUID	NO	
workspace_id	UUID	NO	
role_code	VARCHAR(64)	NO	
is_default	BOOLEAN	NO	false
granted_at	TIMESTAMP	NO	now()
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

No rows in scope (0 total in this filter).

core.user_tenant_access

User ↔ tenant role (MES_RMS, MES_Activation, MES_Admin, …).

Primary key: access_id. Foreign keys: tenant_id → core.tenant(tenant_id); user_id → core.app_user(user_id).

Column	Type	Null	Default
access_id	UUID	NO	gen_random_uuid()
user_id	UUID	NO	
tenant_id	UUID	NO	
role_code	VARCHAR(64)	NO	
is_default	BOOLEAN	NO	false
granted_at	TIMESTAMP	NO	now()
created_at	TIMESTAMP	NO	now()

Column	Type	Null	Default
updated_at	TIMESTAMP	NO	now()

No rows in scope (0 total in this filter).

template.industry_capability

Which platform capabilities each industry template enables.

Primary key: industry_capability_id. Foreign keys: capability_id → core.capability(capability_id); industry_template_id → template.industry_template(industry_template_id).

Column	Type	Null	Default
industry_capability_id	UUID	NO	gen_random_uuid()
industry_template_id	UUID	NO	
capability_id	UUID	NO	
display_order	INTEGER	NO	100
is_required	BOOLEAN	NO	false
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

Data (10 of 10 row(s) in GR_CARBONS / catalog scope):

created_at	metadata	industry_capability_id	industry_template_id	capability_id	display_order	is_required	updated_at
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	10	True	2026-05-31 03:57
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	20	True	2026-05-31 03:57
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	30	True	2026-05-31 03:57
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	40	True	2026-05-31 03:57
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	50	False	2026-05-31 03:57
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	60	True	2026-05-31 03:57
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	70	True	2026-05-31 03:57
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	80	False	2026-05-31 03:57
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	90	True	2026-05-31 03:57
2026-05-31 03:57	{"industry": "ACTIVATED_CARBON"}	f9000000	a1000000	f8000000	100	False	2026-05-31 03:57

template.industry_template

Industry packs (Activated Carbon, Automobile OEM, …).

Primary key: industry_template_id.

Column	Type	Null	Default
industry_template_id	UUID	NO	gen_random_uuid()

Column	Type	Null	Default
template_code	VARCHAR(64)	NO	
template_name	VARCHAR(256)	NO	
description	TEXT	YES	
template_version	VARCHAR(32)	NO	'1.0.0'::character varying
is_active	BOOLEAN	NO	true
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
module_code	VARCHAR(128)	YES	
landing_page	VARCHAR(512)	YES	
ui_module_name	VARCHAR(128)	YES	
api_module_name	VARCHAR(128)	YES	
icon_name	VARCHAR(64)	YES	
dashboard_component	VARCHAR(128)	YES	
display_order	INTEGER	NO	100

Data (2 of 2 row(s) in GR_CARBONS / catalog scope):

template_code	created_at	metadata	industry_template_id	template_name	description	template_version	is_active	updated_at	module_code	landing_page	ui_module_name
ACTIVATED_CARBON	2026-05-31 03:57	{"default_uom": "KG", "process_families": ["Pyrolysis", "Activation", "Screening", "QC"]}	a1000000	Activated Carbon Manufacturing	Kiln activation, screening, quality testing, yield analysis, and traceability for activated carbon plants.	1.0.0	True	2026-05-31 03:57	activated-carbon	/activated-carbon/dashboard	ActivatedCarbonModule
AUTOMOBILE_OEM	2026-05-28 12:09	{"default_uom": "EA", "process_families": ["Casting", "Machining", "Assembly"]}	a1000000	Automobile OEM Supplier	Casting, machining, and assembly workflows for automotive Tier-1/Tier-2 suppliers.	1.0.0	True	2026-05-31 03:57	automotive-oem	/automotive-oem/dashboard	AutomotiveOemModule

template.raw_material_parameter_profile

Shared AC raw-material QC / yield bands (not per-route).

Primary key: id.

Column	Type	Null	Default
id	UUID	NO	gen_random_uuid()
industry_template_code	VARCHAR(64)	NO	
raw_material_type	VARCHAR(128)	NO	
profile_code	VARCHAR(128)	NO	
profile_name	VARCHAR(256)	YES	
parameter_code	VARCHAR(64)	NO	
parameter_name	VARCHAR(256)	YES	
min_value	NUMERIC(18, 6)	YES	

Column	Type	Null	Default
max_value	NUMERIC(18, 6)	YES	
unit	VARCHAR(32)	YES	
expected_yield_percent	NUMERIC(8, 4)	YES	
expected_ctc_min	NUMERIC(18, 6)	YES	
expected_ctc_max	NUMERIC(18, 6)	YES	
expected_density_min	NUMERIC(18, 6)	YES	
expected_density_max	NUMERIC(18, 6)	YES	
is_active	BOOLEAN	NO	true
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NO	CURRENT_TIMESTAMP

Data (21 of 21 row(s) in GR_CARBONS / catalog scope):

created_at	id	industry_template_code	raw_material_type	profile_code	profile_name	parameter_code	parameter_name	min_value	max_value	unit	expected_yield_percent
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	GRANULATED_CHARCOAL	AC-RMP-GRANULATED	Granulated Charcoal	T1_TEMPERATURE	T1 Temperature	450	550	C	22
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	GRANULATED_CHARCOAL	AC-RMP-GRANULATED	Granulated Charcoal	T2_TEMPERATURE	T2 Temperature	550	650	C	22
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	GRANULATED_CHARCOAL	AC-RMP-GRANULATED	Granulated Charcoal	T3_TEMPERATURE	T3 Temperature	650	750	C	22
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	GRANULATED_CHARCOAL	AC-RMP-GRANULATED	Granulated Charcoal	T4_TEMPERATURE	T4 Temperature	750	850	C	22
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	GRANULATED_CHARCOAL	AC-RMP-GRANULATED	Granulated Charcoal	PRESSURE	Pressure	0.5	2.5	bar	22
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	GRANULATED_CHARCOAL	AC-RMP-GRANULATED	Granulated Charcoal	STEAM_FLOW	Steam Flow	200	600	kg/h	22
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	GRANULATED_CHARCOAL	AC-RMP-GRANULATED	Granulated Charcoal	RESIDENCE_TIME	Residence Time	45	120	min	22
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	COCONUT_SHELL_CHARCOAL	AC-RMP-COCO-SHELL	Coconut Shell Charcoal	T1_TEMPERATURE	T1 Temperature	500	600	C	25
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	COCONUT_SHELL_CHARCOAL	AC-RMP-COCO-SHELL	Coconut Shell Charcoal	T2_TEMPERATURE	T2 Temperature	600	700	C	25
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	COCONUT_SHELL_CHARCOAL	AC-RMP-COCO-SHELL	Coconut Shell Charcoal	T3_TEMPERATURE	T3 Temperature	700	800	C	25
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	COCONUT_SHELL_CHARCOAL	AC-RMP-COCO-SHELL	Coconut Shell Charcoal	T4_TEMPERATURE	T4 Temperature	800	900	C	25
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	COCONUT_SHELL_CHARCOAL	AC-RMP-COCO-SHELL	Coconut Shell Charcoal	PRESSURE	Pressure	0.8	3	bar	25
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	COCONUT_SHELL_CHARCOAL	AC-RMP-COCO-SHELL	Coconut Shell Charcoal	STEAM_FLOW	Steam Flow	300	800	kg/h	25
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	COCONUT_SHELL_CHARCOAL	AC-RMP-COCO-SHELL	Coconut Shell Charcoal	RESIDENCE_TIME	Residence Time	60	150	min	25
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	WOOD_CHARCOAL	AC-RMP-WOOD	Wood Charcoal	T1_TEMPERATURE	T1 Temperature	400	500	C	18
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	WOOD_CHARCOAL	AC-RMP-WOOD	Wood Charcoal	T2_TEMPERATURE	T2 Temperature	500	600	C	18
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	WOOD_CHARCOAL	AC-RMP-WOOD	Wood Charcoal	T3_TEMPERATURE	T3 Temperature	600	700	C	18

created_at	id	industry_template_code	raw_material_type	profile_code	profile_name	parameter_code	parameter_name	min_value	max_value	unit	expected_yield_percent
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	WOOD_CHARCOAL	AC-RMP-WOOD	Wood Charcoal	T4_TEMPERATURE	T4 Temperature	700	800	C	18
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	WOOD_CHARCOAL	AC-RMP-WOOD	Wood Charcoal	PRESSURE	Pressure	0.4	2	bar	18
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	WOOD_CHARCOAL	AC-RMP-WOOD	Wood Charcoal	STEAM_FLOW	Steam Flow	150	500	kg/h	18
2026-05-31 13:32	a4000000	ACTIVATED_CARBON	WOOD_CHARCOAL	AC-RMP-WOOD	Wood Charcoal	RESIDENCE_TIME	Residence Time	40	100	min	18

mst.customer

End customers. Also used as the dummy buyer on inward (supplier receipt) POs.

Primary key: customer_id. Foreign keys: tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
customer_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
customer_code	VARCHAR(64)	NO	
customer_name	VARCHAR(256)	NO	
status	VARCHAR(32)	NO	'ACTIVE'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (1 of 1 row(s) in GR_CARBONS / catalog scope):

customer_code	status	customer_name	created_at	metadata	customer_id	updated_at	workspace_id
CUST-AC-001	ACTIVE	Demo Customer	2026-10-02 11:56	{"segment": "Internal", "industry": "ACTIVATED_CARBON"}	c5000000	2026-10-03 12:13	e7000000

mst.equipment

Physical equipment linked to a resource.

Primary key: equipment_id. Foreign keys: resource_id → mst.resource(resource_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
equipment_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
equipment_code	VARCHAR(64)	NO	
equipment_name	VARCHAR(256)	NO	
resource_id	UUID	YES	
equipment_type	VARCHAR(64)	YES	
status	VARCHAR(32)	NO	'ACTIVE'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()

Column	Type	Null	Default
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (3 of 3 row(s) in GR_CARBONS / catalog scope):

equipment_code	status	created_at	metadata	equipment_id	equipment_name	resource_id	equipment_type	updated_at	workspace_id
EQ-KILN-0002	ACTIVE	2026-10-03 18:38	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	e4bb7bfa	Klin 02	0d4428e7	KILN	2026-10-03 18:38	e7000000
EQ-KILN-01	ACTIVE	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "max_temp_c": 950}	32000000	Rotary Activation Kiln 01	31000000	KILN	2026-10-02 11:56	e7000000
EQ-SCREEN-01	ACTIVE	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON"}	32000000	Vibrating Screen 01	31000000	SCREEN	2026-10-02 11:56	e7000000

mst.modbus_register_map

Operational / catalog table.

Primary key: id. Foreign keys: device_id → mst.sensor_device(id); parameter_id → mst.sensor_parameter(id).

Column	Type	Null	Default
id	BIGINT	NO	nextval('mst.modbus_register_map_id_seq…
device_id	BIGINT	NO	
parameter_id	BIGINT	NO	
slave_id	INTEGER	NO	1
function_code	INTEGER	NO	3
register_address	INTEGER	NO	
register_count	INTEGER	NO	2
data_type	VARCHAR(50)	NO	'float32'::character varying
byte_order	VARCHAR(50)	NO	'AB_CD'::character varying
scale_factor	NUMERIC	NO	1
offset_value	NUMERIC	NO	0
simulation_min	NUMERIC	YES	
simulation_max	NUMERIC	YES	
polling_interval_sec	INTEGER	NO	60
is_active	BOOLEAN	NO	true
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

Data (7 of 7 row(s) in GR_CARBONS / catalog scope):

created_at	id	device_id	parameter_id	slave_id	function_code	register_address	register_count	data_type	byte_order	scale_factor	offset_value
2026-06-06 23:13	1	1	1	1	3	40001	2	float32	AB_CD	1	0
2026-06-06 23:13	2	1	2	1	3	40003	2	float32	AB_CD	1	0
2026-06-06 23:13	3	1	3	1	3	40005	2	float32	AB_CD	1	0
2026-06-06 23:13	4	1	4	1	3	40007	2	float32	AB_CD	1	0
2026-06-06 23:13	5	1	5	1	3	40009	2	float32	AB_CD	1	0

created_at	id	device_id	parameter_id	slave_id	function_code	register_address	register_count	data_type	byte_order	scale_factor	offset_value
2026-06-06 23:13	6	1	6	1	3	40011	2	float32	AB_CD	1	0
2026-06-06 23:13	7	1	7	1	3	40013	2	float32	AB_CD	1	0

mst.operation_resource_requirement

Which resource an operation needs.

Primary key: requirement_id. Foreign keys: resource_id → mst.resource(resource_id); routing_operation_id → mst.routing_operation(routing_operation_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
requirement_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
routing_operation_id	UUID	NO	
resource_id	UUID	NO	
requirement_type	VARCHAR(32)	NO	'PRIMARY'::character varying
quantity	NUMERIC(18, 4)	NO	1
uom	VARCHAR(16)	NO	'EA'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (11 of 11 row(s) in GR_CARBONS / catalog scope):

quantity	created_at	metadata	requirement_id	routing_operation_id	resource_id	requirement_type	uom	updated_at	workspace_id
1	2026-10-02 11:56	{"process_step": "PYROLYSIS"}	35000000	34000000	31000000	PRIMARY	EA	2026-10-02 11:56	
2	2026-10-02 11:56	{"operators": 2}	35000000	34000000	31000000	LABOR	EA	2026-10-02 11:56	
1	2026-10-02 11:56	{}	35000000	34000000	31000000	TOOL	EA	2026-10-02 11:56	
1	2026-10-02 11:56	{"process_step": "ACTIVATION"}	35000000	34000000	31000000	PRIMARY	EA	2026-10-02 11:56	
2	2026-10-02 11:56	{"operators": 2}	35000000	34000000	31000000	LABOR	EA	2026-10-02 11:56	
1	2026-10-02 11:56	{}	35000000	34000000	31000000	TOOL	EA	2026-10-02 11:56	
1	2026-10-02 11:56	{"process_step": "SCREENING"}	35000000	34000000	31000000	PRIMARY	EA	2026-10-02 11:56	
1	2026-10-02 11:56	{"operators": 1}	35000000	34000000	31000000	LABOR	EA	2026-10-02 11:56	
1	2026-10-02 11:56	{}	35000000	34000000	31000000	TOOL	EA	2026-10-02 11:56	
1	2026-10-02 11:56	{"operators": 1}	35000000	34000000	31000000	LABOR	EA	2026-10-02 11:56	
1	2026-10-02 11:56	{}	35000000	34000000	31000000	TOOL	EA	2026-10-02 11:56	

mst.parameter_business_mapping

Operational / catalog table.

Primary key: id. Foreign keys: device_id → mst.sensor_device(id); parameter_id → mst.sensor_parameter(id).

Column	Type	Null	Default
id	BIGINT	NO	nextval('mst.parameter_business_mapping…

Column	Type	Null	Default
industry_code	VARCHAR(100)	NO	
device_id	BIGINT	NO	
parameter_id	BIGINT	NO	
target_entity	VARCHAR(100)	NO	
target_field	VARCHAR(100)	NO	
process_step_code	VARCHAR(100)	YES	
work_center_code	VARCHAR(100)	YES	
equipment_code	VARCHAR(100)	YES	
is_active	BOOLEAN	NO	true
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

Data (7 of 7 row(s) in GR_CARBONS / catalog scope):

equipment_code	created_at	id	industry_code	device_id	parameter_id	target_entity	target_field	process_step_code	work_center_code	is_active	updated_at
KILN_01	2026-06-06 23:13	1	activated_carbon	1	1	kiln	temperature_zone_1		KILN_01	True	2026-06-06 23:13
KILN_01	2026-06-06 23:13	2	activated_carbon	1	2	kiln	temperature_zone_2		KILN_01	True	2026-06-06 23:13
KILN_01	2026-06-06 23:13	3	activated_carbon	1	3	kiln	temperature_zone_3		KILN_01	True	2026-06-06 23:13
KILN_01	2026-06-06 23:13	4	activated_carbon	1	4	kiln	temperature_zone_4		KILN_01	True	2026-06-06 23:13
KILN_01	2026-06-06 23:13	5	activated_carbon	1	5	kiln	pressure		KILN_01	True	2026-06-06 23:13
KILN_01	2026-06-06 23:13	6	activated_carbon	1	6	kiln	steam_flow		KILN_01	True	2026-06-06 23:13
KILN_01	2026-06-06 23:13	7	activated_carbon	1	7	kiln	residence_time		KILN_01	True	2026-06-06 23:13

mst.product

Finished / intermediate grades (FG-AC-*).

Primary key: product_id. Foreign keys: tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
product_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
product_code	VARCHAR(64)	NO	
product_name	VARCHAR(256)	NO	
product_type	VARCHAR(64)	YES	
uom	VARCHAR(16)	NO	'EA'::character varying
status	VARCHAR(32)	NO	'ACTIVE'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (3 of 3 row(s) in GR_CARBONS / catalog scope):

product_code	status	product_name	created_at	metadata	product_id	product_type	uom	updated_at	workspace_id
FG-AC-1	ACTIVE	GR Granular Activated Carbon	2026-10-02 12:05	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	50bfcbfa	FINISHED_GOOD	KG	2026-10-02 12:05	e7000000
AC-GRAN-8x30	ACTIVE	Granular Activated Carbon 8x30	2026-10-02 11:56	{"mesh": "8x30", "industry": "ACTIVATED_CARBON", "iodine_target_mg_g": 950}	a6000000	FINISHED	KG	2026-10-02 11:56	e7000000
RAW-COCO-SHELL	ACTIVE	Coconut Shell (GR)	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "operations": [{"is_mandatory": true, "process_step": "KILN_FEED", "operation_code": "OP-KILN-FEED", "operation_name": "P2-2 Kiln feed \u2014 load bag & charge weight", "operation_sequen…	a6000000	RAW_MATERIAL	KG	2026-10-04 19:11	e7000000

mst.resource

Machines, rotaries, kilns, labor, material resources.

Primary key: resource_id. Foreign keys: tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
resource_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
resource_code	VARCHAR(64)	NO	
resource_name	VARCHAR(256)	NO	
resource_type	VARCHAR(64)	NO	
status	VARCHAR(32)	NO	'ACTIVE'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (11 of 11 row(s) in GR_CARBONS / catalog scope):

resource_code	status	resource_name	created_at	metadata	resource_id	resource_type	updated_at	workspace_id
RES-POSTACT-01	ACTIVE	Post activation	2026-10-04 11:57	{"role": "post_activation", "stations": {"fg_qc": {"load_events": [], "loaded_bags": [], "output_bags": []}, "blending": {"load_events": [], "loaded_bags": [], "output_bags": []}, "crushing": {"load_events": [], "loaded…	57261196	MACHINE	2026-10-05 05:53	e7000000

resource_code	status	resource_name	created_at	metadata	resource_id	resource_type	updated_at	workspace_id
RES-DESTON-01	ACTIVE	De-Stoner	2026-10-04 05:20	{"role": "destoning", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "destoning_load_events": [{"kg": 200.0, "bag_code": "KA_041026_00002", "lot_code": "LOT-AC-6", "loaded_at": "2026-10-04T08:11:40.211329+00:00", "kil…	3c018e7d	MACHINE	2026-10-05 05:53	e7000000
RES-KILN-0002	IN_USE	Klin 02	2026-10-03 18:38	{"role": "kiln", "loaded_kg": "22894.0", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "active_lot_code": "LOT-AC-23", "equipment_class": "KILN", "kiln_output_seq": 12, "active_lot_codes": ["LOT-AC-11", "LOT-AC-2", "…	0d4428e7	MACHINE	2026-10-04 20:55	e7000000
RES-TOOL-CHARGER	ACTIVE	Kiln Charging Hopper	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "process_step": "PYROLYSIS"}	31000000	TOOL	2026-10-02 11:56	e7000000
RES-TOOL-STEAM	ACTIVE	Steam Injection Lance	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "process_step": "ACTIVATION"}	31000000	TOOL	2026-10-02 11:56	e7000000
RES-SCREEN-01	ACTIVE	Vibrating Screen Line 1	2026-10-02 11:56	{"mesh": "8x30", "industry": "ACTIVATED_CARBON"}	31000000	MACHINE	2026-10-02 11:56	e7000000
RES-TOOL-LAB-KIT	ACTIVE	Iodine & Moisture Test Kit	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "process_step": "QC"}	31000000	TOOL	2026-10-02 11:56	e7000000
RES-KILN-01	AVAILABLE	Activation Kiln 01	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "loaded_kg": "23276.0", "capacity_mt": 5, "kiln_output_seq": 2, "active_lot_codes": ["LOT-AC-10", "LOT-AC-12", "LOT-AC-3", "LOT-AC-6", "LOT-AC-4", "LOT-AC-7"], "kiln_feed_events": [{"kg"…	31000000	MACHINE	2026-10-04 08:34	e7000000
RES-TOOL-SIEVE-8x30	ACTIVE	8x30 Classification Sieve Set	2026-10-02 11:56	{"mesh": "8x30", "industry": "ACTIVATED_CARBON", "process_step": "SCREENING"}	31000000	TOOL	2026-10-02 11:56	e7000000
RES-LAB-QC	ACTIVE	Quality Lab Team	2026-10-02 11:56	{"shift": "A", "industry": "ACTIVATED_CARBON", "operators": 2}	31000000	LABOR	2026-10-02 11:56	e7000000
RES-LAB-AC-01	ACTIVE	Activation Shift Crew	2026-10-02 11:56	{"shift": "A", "industry": "ACTIVATED_CARBON", "operators": 4}	31000000	LABOR	2026-10-02 11:56	e7000000

mst.routing

Process route for a product (AC standard process).

Primary key: routing_id. Foreign keys: product_id → mst.product(product_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
routing_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
routing_code	VARCHAR(64)	NO	
product_id	UUID	YES	
routing_version	VARCHAR(32)	NO	'1'::character varying
process_family	VARCHAR(64)	YES	
is_active	BOOLEAN	NO	true
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	
industry_template_code	VARCHAR(64)	YES	
source_template_code	VARCHAR(64)	YES	
is_template_route	BOOLEAN	NO	false
is_template_copy	BOOLEAN	NO	false
is_customized	BOOLEAN	NO	false

Data (2 of 2 row(s) in GR_CARBONS / catalog scope):

routing_code	created_at	metadata	routing_id	product_id	routing_version	process_family	is_active	updated_at	workspace_id	industry_template_code	source_template_code
RT-AC-2	2026-10-02 12:05	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	8b1ea279	50bfcbfa	1	ACTIVATED_CARBON	True	2026-10-02 12:05	e7000000		
RT-AC-8x30-V1	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON"}	33000000	a6000000	1	Activation	True	2026-10-02 11:56	e7000000		

mst.routing_operation

Ordered operations on a route (kiln feed, destoning, screening, …).

Primary key: routing_operation_id. Foreign keys: routing_id → mst.routing(routing_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
routing_operation_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
routing_id	UUID	NO	
operation_code	VARCHAR(64)	NO	
operation_sequence	INTEGER	NO	
operation_name	VARCHAR(256)	NO	

Column	Type	Null	Default
standard_cycle_min	NUMERIC(12, 4)	YES	
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	
industry_template_code	VARCHAR(64)	YES	
source_template_operation_code	VARCHAR(64)	YES	
operation_seq	INTEGER	YES	
operation_type	VARCHAR(64)	YES	
is_quality_gate	BOOLEAN	NO	false
is_feasibility_gate	BOOLEAN	NO	false
is_active	BOOLEAN	NO	true
is_customized	BOOLEAN	NO	false

Data (16 of 16 row(s) in GR_CARBONS / catalog scope):

operation_code	operation_name	created_at	metadata	routing_operation_id	routing_id	operation_sequence	standard_cycle_min	updated_at	workspace_id	industry_template_code	source_template_operation_code
OP-KILN-FEED	P2-2 Kiln feed — load bag & charge weight	2026-10-02 12:05	{"is_mandatory": true, "process_code": "KILN_FEED", "process_step": "KILN_FEED", "process_type": "KILN_FEED", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "ACT…	caf7b3c2	8b1ea279	1		2026-10-04 08:55	e7000000		
OP-DEDUST	P4-3 De-dusting — screening bin → de-duster, grade & wt	2026-10-02 12:05	{"is_mandatory": true, "process_code": "DRYING", "process_step": "DRYING", "process_type": "DRYING", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "ACTIVATED_CA…	9cc0842f	8b1ea279	7		2026-10-04 18:16	e7000000		

operation_code	operation_name	created_at	metadata	routing_operation_id	routing_id	operation_sequence	standard_cycle_min	updated_at	workspace_id	industry_template_code	source_template_operation_code
OP-DESTONING	P3-5 Destoning — kiln O/P bag → destoning loader	2026-10-02 12:05	{"is_mandatory": true, "process_code": "DESTONING", "process_step": "DESTONING", "process_type": "DESTONING", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "ACT…	a25971a3	8b1ea279	4		2026-10-04 08:55	e7000000		
OP-POST-SCREEN	P3 Post-activation screening (+4, 4x6, 6x8, -8, MBD, CBD)	2026-10-02 12:05	{"is_mandatory": true, "process_code": "SCREENING", "process_step": "SCREENING", "process_type": "SCREENING", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "ACT…	ab8f805e	8b1ea279	6		2026-10-04 08:55	e7000000		
OP-KILN-FEED-QC	P2-1 Kiln feed bag quality (+2…-14, moisture, dust, volatile)	2026-10-02 12:05	{"is_mandatory": true, "process_code": "KILN_FEED", "process_step": "KILN_FEED", "process_type": "KILN_FEED", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "ACT…	cab75d16	8b1ea279	2		2026-10-04 08:55	e7000000		

operation_code	operation_name	created_at	metadata	routing_operation_id	routing_id	operation_sequence	standard_cycle_min	updated_at	workspace_id	industry_template_code	source_template_operation_code
OP-KILN-OUT-QC	P2-3 Kiln output quality (+3…-30, CBD, etc.)	2026-10-02 12:05	{"is_mandatory": true, "process_code": "ACTIVATION", "process_step": "ACTIVATION", "process_type": "ACTIVATION", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "…	64d837bc	8b1ea279	3		2026-10-04 08:55	e7000000		
OP-FG-QC	Final — finished goods QC (after post-activation screening)	2026-10-02 12:05	{"is_mandatory": true, "process_code": "QC", "process_step": "QC", "process_type": "QC", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "ACTIVATED_CARBON"}	718a96c8	8b1ea279	12		2026-10-04 08:55	e7000000		
OP-KILN-TEMP	P2-4 Kiln temperature & telemetry (T1–T4, RPM, dampers, steam)	2026-10-02 12:05	{"is_mandatory": true, "process_code": "ACTIVATION", "process_step": "ACTIVATION", "process_type": "ACTIVATION", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "…	7b6628b6	8b1ea279	11		2026-10-04 08:55	e7000000		

operation_code	operation_name	created_at	metadata	routing_operation_id	routing_id	operation_sequence	standard_cycle_min	updated_at	workspace_id	industry_template_code	source_template_operation_code
OP-DESTON-QC	P3-6 Destoning bag quality (+3…-30, CBD, etc.)	2026-10-02 12:05	{"is_mandatory": true, "process_code": "SCREENING", "process_step": "SCREENING", "process_type": "SCREENING", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "ACT…	86967ce3	8b1ea279	5		2026-10-04 08:55	e7000000		
OP-CRUSH-LOAD	P4-2 Crushing — input bag → crusher output grade	2026-10-02 12:05	{"is_mandatory": true, "process_code": "GRINDING", "process_step": "GRINDING", "process_type": "GRINDING", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "ACTIVA…	067fdcfe	8b1ea279	8		2026-10-04 18:16	e7000000		
OP-DEMAG	P4-4 De-magnetisation — bag load → grade out	2026-10-02 12:05	{"is_mandatory": true, "process_code": "GRINDING", "process_step": "GRINDING", "process_type": "GRINDING", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry_template_code": "ACTIVA…	0c7bf68a	8b1ea279	9		2026-10-04 08:55	e7000000		

operation_code	operation_name	created_at	metadata	routing_operation_id	routing_id	operation_sequence	standard_cycle_min	updated_at	workspace_id	industry_template_code	source_template_operation_code
OP-BLEND	P4-5 Blending — bag load → output grade	2026-10-02 12:05	{"is_mandatory": true, "process_code": "GRADE_SEPARATION", "process_step": "GRADE_SEPARATION", "process_type": "GRADE_SEPARATION", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "inspection_required": false, "industry…	63f60409	8b1ea279	10		2026-10-04 08:55	e7000000		
OP-ACTIVATION	Steam Activation	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "process_step": "ACTIVATION"}	34000000	33000000	20	480	2026-10-02 11:56	e7000000		
OP-SCREENING	Screening & Classification	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "process_step": "SCREENING"}	34000000	33000000	30	120	2026-10-02 11:56	e7000000		
OP-QC	Quality Testing	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "process_step": "QC"}	34000000	33000000	40	60	2026-10-02 11:56	e7000000		
OP-PYROLYSIS	Pyrolysis / Carbonization	2026-10-02 11:56	{"industry": "ACTIVATED_CARBON", "process_step": "PYROLYSIS"}	34000000	33000000	10	360	2026-10-02 11:56	e7000000		

mst.routing_operation_parameter

Per-operation capture parameters and limits.

Primary key: id. Foreign keys: routing_operation_id → mst.routing_operation(routing_operation_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
workspace_id	UUID	YES	
routing_operation_id	UUID	NO	
parameter_code	VARCHAR(64)	YES	
parameter_name	VARCHAR(256)	YES	
data_type	VARCHAR(32)	YES	
unit	VARCHAR(32)	YES	
min_value	NUMERIC(18, 6)	YES	
max_value	NUMERIC(18, 6)	YES	
is_mandatory	BOOLEAN	NO	false

Column	Type	Null	Default
reading_frequency	VARCHAR(64)	YES	
display_order	INTEGER	YES	
is_active	BOOLEAN	NO	true
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NO	CURRENT_TIMESTAMP

No rows in scope (0 total in this filter).

mst.sensor_device

Operational / catalog table.

Primary key: id.

Column	Type	Null	Default
id	BIGINT	NO	nextval('mst.sensor_device_id_seq'::reg…
device_code	VARCHAR(100)	NO	
device_name	VARCHAR(200)	NO	
industry_code	VARCHAR(100)	NO	
site_code	VARCHAR(100)	YES	
area_code	VARCHAR(100)	YES	
work_center_code	VARCHAR(100)	YES	
equipment_code	VARCHAR(100)	YES	
protocol	VARCHAR(50)	NO	'MODBUS_TCP'::character varying
connection_type	VARCHAR(50)	NO	'SIMULATION'::character varying
ip_address	VARCHAR(100)	YES	
port	INTEGER	NO	502
serial_port	VARCHAR(100)	YES	
baud_rate	INTEGER	YES	
parity	VARCHAR(20)	YES	
stop_bits	INTEGER	YES	
is_active	BOOLEAN	NO	true
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

Data (1 of 1 row(s) in GR_CARBONS / catalog scope):

equipment_code	created_at	id	device_code	device_name	industry_code	site_code	area_code	work_center_code	protocol	connection_type	ip_address
KILN_01	2026-06-06 23:13	1	KILN_01_MODBUS_SIM	Kiln 01 Smart Modbus Simulator	activated_carbon			KILN_01	MODBUS_TCP	SIMULATION	

mst.sensor_parameter

Operational / catalog table.

Primary key: id.

Column	Type	Null	Default
id	BIGINT	NO	nextval('mst.sensor_parameter_id_seq'::…

Column	Type	Null	Default
parameter_code	VARCHAR(100)	NO	
parameter_name	VARCHAR(200)	NO	
data_type	VARCHAR(50)	NO	'number'::character varying
uom	VARCHAR(50)	YES	
normal_min	NUMERIC	YES	
normal_max	NUMERIC	YES	
is_active	BOOLEAN	NO	true
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

Data (7 of 7 row(s) in GR_CARBONS / catalog scope):

created_at	id	parameter_code	parameter_name	data_type	uom	normal_min	normal_max	is_active	updated_at
2026-06-06 23:13	1	TEMP_ZONE_1	Temperature Zone 1	number	C	750	950	True	2026-06-06 23:13
2026-06-06 23:13	2	TEMP_ZONE_2	Temperature Zone 2	number	C	750	950	True	2026-06-06 23:13
2026-06-06 23:13	3	TEMP_ZONE_3	Temperature Zone 3	number	C	750	950	True	2026-06-06 23:13
2026-06-06 23:13	4	TEMP_ZONE_4	Temperature Zone 4	number	C	750	950	True	2026-06-06 23:13
2026-06-06 23:13	5	PRESSURE	Pressure	number	bar	0.8	1.5	True	2026-06-06 23:13
2026-06-06 23:13	6	STEAM_FLOW	Steam Flow	number	kg/hr	100	300	True	2026-06-06 23:13
2026-06-06 23:13	7	RESIDENCE_TIME	Residence Time	number	hr	20	40	True	2026-06-06 23:13

mst.supplier

Raw-material vendors for gate receipt.

Primary key: supplier_id. Foreign keys: tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
supplier_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
workspace_id	UUID	YES	
supplier_code	VARCHAR(64)	NO	
supplier_name	VARCHAR(256)	NO	
status	VARCHAR(32)	NO	'ACTIVE'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()

Data (1 of 1 row(s) in GR_CARBONS / catalog scope):

supplier_code	status	supplier_name	created_at	metadata	supplier_id	workspace_id	updated_at
SUP-AC-1	ACTIVE	GR Smoke Supplier	2026-10-02 13:07	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	159110d0	e7000000	2026-10-02 13:07

trn.customer_order

Inward receipts (ORD-AC-*) and customer sales POs (CUST_PO_*). Shop-floor RM state lives in metadata JSON.

Primary key: customer_order_id. Foreign keys: customer_id → mst.customer(customer_id); supplier_id → mst.supplier(supplier_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
customer_order_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
order_code	VARCHAR(64)	NO	
customer_id	UUID	YES	
order_date	DATE	NO	CURRENT_DATE
status	VARCHAR(32)	NO	'OPEN'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	
supplier_id	UUID	YES	

Data (24 of 24 row(s) in GR_CARBONS / catalog scope):

order_code	status	order_date	created_at	metadata	customer_order_id	customer_id	updated_at	workspace_id	supplier_id
ORD-AC-22	OPEN	2026-10-05	2026-10-05 06:57	{"flow": "RAW_MATERIAL_INWARD", "inward_receipt": {"dc_number": "DC5678", "our_weight_kg": "2300", "supplier_weight_kg": "2300"}, "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_A…	30070574		2026-10-05 06:57	e7000000	159110d0
CUST_PO_5	CLOSED	2026-10-05	2026-10-05 05:38	{"flow": "CUSTOMER_SALES", "ordered_kg": "1000", "product_id": "50bfcbfa-1ad1-4706-bbe8-b6e6ddecd83d", "dispatched_kg": "1000", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "last_dispatch_id": "DISP-AC-1", "last_dis…	f26b7b5f	c5000000	2026-10-05 05:53	e7000000	
CUST_PO_4	OPEN	2026-10-05	2026-10-05 05:21	{"flow": "CUSTOMER_SALES", "ordered_kg": "1000", "product_id": "50bfcbfa-1ad1-4706-bbe8-b6e6ddecd83d", "dispatched_kg": "0", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	24d80e64	c5000000	2026-10-05 05:48	e7000000	

order_code	status	order_date	created_at	metadata	customer_order_id	customer_id	updated_at	workspace_id	supplier_id
ORD-AC-21	OPEN	2026-10-05	2026-10-05 02:54	{"flow": "RAW_MATERIAL_INWARD", "inward_receipt": {"dc_number": "DC12333", "our_weight_kg": "2500", "supplier_weight_kg": "2500"}, "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_…	d06819bc		2026-10-05 02:55	e7000000	159110d0
ORD-AC-20	QC_PASSED	2026-10-04	2026-10-04 20:47	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-04T20:49:26.259760+00:00", "source": "inbound_execution_lab", "remarks": null, "ad_value": "4", "dust_pct": "4", "qc_status": "PASS", "moisture_pct": "4"}…	63d91e23		2026-10-04 20:55	e7000000	159110d0
ORD-AC-19	QC_PASSED	2026-10-04	2026-10-04 20:41	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-04T20:41:30.092612+00:00", "source": "inbound_execution_lab", "remarks": null, "ad_value": "22", "dust_pct": "1", "qc_status": "PASS", "moisture_pct": "1"…	064dd858		2026-10-04 20:44	e7000000	159110d0
ORD-AC-18	OPEN	2026-10-04	2026-10-04 20:32	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-04T20:33:34.272803+00:00", "source": "inbound_execution_lab", "remarks": null, "ad_value": "1222", "dust_pct": "1", "qc_status": "PASS", "moisture_pct": "…	a4704c3b		2026-10-04 20:34	e7000000	159110d0
ORD-AC-17	QC_PASSED	2026-10-04	2026-10-04 18:41	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-04T18:43:00.386141+00:00", "source": "inbound_execution_lab", "remarks": null, "ad_value": "1", "dust_pct": "1", "qc_status": "PASS", "moisture_pct": "1"}…	84d26087		2026-10-04 20:01	e7000000	159110d0

order_code	status	order_date	created_at	metadata	customer_order_id	customer_id	updated_at	workspace_id	supplier_id
ORD-AC-16	OPEN	2026-10-04	2026-10-04 18:33	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-04T18:34:34.830193+00:00", "source": "inbound_execution_lab", "remarks": null, "ad_value": "10", "dust_pct": "1", "qc_status": "PASS", "moisture_pct": "2"…	84b8a9db		2026-10-04 19:05	e7000000	159110d0
ORD-AC-15	OPEN	2026-10-04	2026-10-04 18:23	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-04T18:24:42.442557+00:00", "source": "inbound_execution_lab", "ad_value": "12222", "dust_pct": "12", "qc_status": "FAIL", "moisture_pct": "12"}, "inward_r…	02cc9df2		2026-10-04 18:24	e7000000	159110d0
ORD-AC-14	OPEN	2026-10-04	2026-10-04 17:31	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-04T17:34:01.311908+00:00", "source": "inbound_execution_lab", "ad_value": "1", "dust_pct": "1", "qc_status": "PASS", "moisture_pct": "1"}, "inward_bags": …	d31d704d		2026-10-04 17:34	e7000000	159110d0
ORD-AC-13	QC_PASSED	2026-10-04	2026-10-04 16:49	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-04T16:50:57.790702+00:00", "source": "inbound_execution_lab", "ad_value": "89", "dust_pct": "20", "qc_status": "PASS", "moisture_pct": "10"}, "inward_bags…	989d5d6d		2026-10-04 20:02	e7000000	159110d0
CUST_PO_3	OPEN	2026-10-04	2026-10-04 16:40	{"flow": "CUSTOMER_SALES", "ordered_kg": "1000", "product_id": "50bfcbfa-1ad1-4706-bbe8-b6e6ddecd83d", "dispatched_kg": "0", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	9317ce85	c5000000	2026-10-05 05:48	e7000000	

order_code	status	order_date	created_at	metadata	customer_order_id	customer_id	updated_at	workspace_id	supplier_id
CUST_PO_1	OPEN	2026-10-04	2026-10-04 16:30	{"flow": "CUSTOMER_SALES", "ordered_kg": "1000", "product_id": "50bfcbfa-1ad1-4706-bbe8-b6e6ddecd83d", "dispatched_kg": "0", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	80819eed	c5000000	2026-10-05 05:48	e7000000	
ORD-AC-10	QC_PASSED	2026-10-04	2026-10-04 15:57	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-04T15:58:44.278528+00:00", "source": "inbound_execution_lab", "ad_value": "85", "dust_pct": "20", "qc_status": "PASS", "moisture_pct": "10"}, "inward_bags…	39656309		2026-10-04 16:05	e7000000	159110d0
CUST_PO_2	OPEN	2026-10-04	2026-10-04 13:27	{"flow": "CUSTOMER_SALES", "ordered_kg": "1000", "product_id": "a6000000-0000-4000-8000-000000000312", "dispatched_kg": "0", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	623a25f6	c5000000	2026-10-05 05:48	e7000000	
ORD-AC-8	QC_PASSED	2026-10-03	2026-10-03 18:06	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-03T18:07:22.725099+00:00", "source": "inbound_execution_lab", "ad_value": "10", "dust_pct": "1", "qc_status": "PASS", "moisture_pct": "6"}, "inward_bags":…	fc283675		2026-10-04 09:35	e7000000	159110d0
ORD-AC-7	OPEN	2026-10-03	2026-10-03 17:40	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-03T18:14:00.224844+00:00", "source": "inbound_execution_lab", "ad_value": "8", "dust_pct": "3", "qc_status": "FAIL", "moisture_pct": "9"}, "inward_receipt…	0a55a160		2026-10-03 18:14	e7000000	159110d0

order_code	status	order_date	created_at	metadata	customer_order_id	customer_id	updated_at	workspace_id	supplier_id
ORD-AC-6	QC_PASSED	2026-10-03	2026-10-03 17:32	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-03T17:32:43.919052+00:00", "source": "inbound_execution_lab", "ad_value": "12", "dust_pct": "2", "qc_status": "PASS", "moisture_pct": "8"}, "inward_bags":…	02d17a82		2026-10-03 17:36	e7000000	159110d0
ORD-AC-5	QC_PASSED	2026-10-03	2026-10-03 17:21	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-03T17:21:56.834695+00:00", "source": "inbound_execution_lab", "ad_value": "40", "dust_pct": "20", "qc_status": "PASS", "moisture_pct": "10"}, "inward_bags…	630a8609		2026-10-03 17:25	e7000000	159110d0
ORD-AC-4	QC_PASSED	2026-10-03	2026-10-03 17:12	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-03T17:13:52.190107+00:00", "source": "inbound_execution_lab", "ad_value": "55", "dust_pct": "20", "qc_status": "PASS", "moisture_pct": "10"}, "inward_bags…	93111bd1		2026-10-03 17:19	e7000000	159110d0
ORD-AC-3	QC_PASSED	2026-10-03	2026-10-03 15:37	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-03T15:39:46.564499+00:00", "source": "inbound_execution_lab", "ad_value": "5555", "dust_pct": "2", "qc_status": "PASS", "moisture_pct": "2"}, "inward_bags…	64cd6b87		2026-10-03 15:49	e7000000	159110d0
ORD-AC-2	QC_PASSED	2026-10-03	2026-10-03 14:54	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-03T14:55:44.092997+00:00", "source": "inbound_execution_lab", "ad_value": "10", "dust_pct": "1", "qc_status": "PASS", "moisture_pct": "2"}, "inward_bags":…	b86353af		2026-10-03 15:09	e7000000	159110d0

order_code	status	order_date	created_at	metadata	customer_order_id	customer_id	updated_at	workspace_id	supplier_id
ORD-AC-1	QC_PASSED	2026-10-03	2026-10-03 12:23	{"flow": "RAW_MATERIAL_INWARD", "initial_qc": {"qc_at": "2026-10-03T13:52:19.992775+00:00", "source": "inbound_execution_lab", "ad_value": "1234", "dust_pct": "2", "qc_status": "PASS", "moisture_pct": "1"}, "inward_bags…	38034fe6		2026-10-03 14:30	e7000000	159110d0

trn.customer_order_line

Order lines (product + qty) for inward and sales.

Primary key: order_line_id. Foreign keys: order_id → trn.customer_order(customer_order_id); product_id → mst.product(product_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
order_line_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
order_id	UUID	NO	
line_code	VARCHAR(64)	NO	
product_id	UUID	NO	
quantity	NUMERIC(18, 4)	NO	
uom	VARCHAR(16)	NO	'EA'::character varying
status	VARCHAR(32)	NO	'OPEN'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (34 of 34 row(s) in GR_CARBONS / catalog scope):

line_code	status	quantity	created_at	metadata	order_line_id	order_id	product_id	uom	updated_at
ORDL-AC-34	OPEN	2300	2026-10-05 06:57	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	2bb45da9	30070574	a6000000	KG	2026-10-05 06:57
ORDL-AC-33	OPEN	1000	2026-10-05 05:38	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	33ee580b	f26b7b5f	50bfcbfa	KG	2026-10-05 05:38
ORDL-AC-32	OPEN	1000	2026-10-05 05:21	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	9373cc06	24d80e64	50bfcbfa	KG	2026-10-05 05:21

line_code	status	quantity	created_at	metadata	order_line_id	order_id	product_id	uom	updated_at
ORDL-AC-31	OPEN	2500	2026-10-05 02:54	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	40e14d98	d06819bc	a6000000	KG	2026-10-05 02:54
ORDL-AC-30	OPEN	3750	2026-10-04 20:52	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	7e4e3d14	63d91e23	50bfcbfa	KG	2026-10-04 20:52
ORDL-AC-29	OPEN	4000	2026-10-04 20:47	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	2e9d1955	63d91e23	a6000000	KG	2026-10-04 20:47
ORDL-AC-28	OPEN	3333	2026-10-04 20:41	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	8a64ed0e	064dd858	a6000000	KG	2026-10-04 20:41
ORDL-AC-27	OPEN	9999	2026-10-04 20:32	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	375fbb23	a4704c3b	a6000000	KG	2026-10-04 20:32
ORDL-AC-26	OPEN	7500	2026-10-04 19:24	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	f7ac7fa2	84d26087	50bfcbfa	KG	2026-10-04 19:24
ORDL-AC-25	OPEN	7500	2026-10-04 18:41	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	4eb9466e	84d26087	a6000000	KG	2026-10-04 18:41
ORDL-AC-24	OPEN	1500	2026-10-04 18:33	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	37e2dec6	84b8a9db	a6000000	KG	2026-10-04 18:33
ORDL-AC-23	OPEN	19999	2026-10-04 18:23	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	5fb08ffc	02cc9df2	a6000000	KG	2026-10-04 18:23
ORDL-AC-22	OPEN	1234	2026-10-04 17:31	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	85aac746	d31d704d	a6000000	KG	2026-10-04 17:31
ORDL-AC-21	OPEN	5555	2026-10-04 17:00	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	ceb69812	989d5d6d	50bfcbfa	KG	2026-10-04 17:00

line_code	status	quantity	created_at	metadata	order_line_id	order_id	product_id	uom	updated_at
ORDL-AC-20	OPEN	5555	2026-10-04 16:49	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	672e8522	989d5d6d	a6000000	KG	2026-10-04 16:49
ORDL-AC-19	OPEN	1000	2026-10-04 16:40	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	3e7a684a	9317ce85	50bfcbfa	KG	2026-10-04 16:40
ORDL-AC-18	OPEN	1000	2026-10-04 16:30	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	34a4d0c5	80819eed	50bfcbfa	KG	2026-10-04 16:30
ORDL-AC-17	OPEN	999	2026-10-04 16:05	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	829fcc44	39656309	50bfcbfa	KG	2026-10-04 16:05
ORDL-AC-16	OPEN	9999	2026-10-04 15:57	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	4fad720d	39656309	a6000000	KG	2026-10-04 15:57
ORDL-AC-15	OPEN	1000	2026-10-04 13:27	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	f9888427	623a25f6	a6000000	KG	2026-10-04 13:27
ORDL-AC-14	OPEN	300	2026-10-03 18:06	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	5cf6114f	fc283675	a6000000	KG	2026-10-03 18:06
ORDL-AC-13	OPEN	480	2026-10-03 17:40	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	ad8ed88d	0a55a160	a6000000	KG	2026-10-03 17:40
ORDL-AC-12	OPEN	1000	2026-10-03 17:36	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	63003eeb	02d17a82	50bfcbfa	KG	2026-10-03 17:36
ORDL-AC-11	OPEN	1000	2026-10-03 17:32	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	c10796af	02d17a82	a6000000	KG	2026-10-03 17:32
ORDL-AC-10	OPEN	9999	2026-10-03 17:25	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	74db2b9b	630a8609	50bfcbfa	KG	2026-10-03 17:25

line_code	status	quantity	created_at	metadata	order_line_id	order_id	product_id	uom	updated_at
ORDL-AC-9	OPEN	9999	2026-10-03 17:21	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	a1750413	630a8609	a6000000	KG	2026-10-03 17:21
ORDL-AC-8	OPEN	4999	2026-10-03 17:18	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	b1618bc6	93111bd1	50bfcbfa	KG	2026-10-03 17:19
ORDL-AC-7	OPEN	9999	2026-10-03 17:12	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	97535809	93111bd1	a6000000	KG	2026-10-03 17:12
ORDL-AC-6	OPEN	7777	2026-10-03 15:48	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	85f957c8	64cd6b87	50bfcbfa	KG	2026-10-03 15:48
ORDL-AC-5	OPEN	7777	2026-10-03 15:37	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	c45c7958	64cd6b87	a6000000	KG	2026-10-03 15:37
ORDL-AC-4	OPEN	5000	2026-10-03 15:08	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	3a1011ee	b86353af	50bfcbfa	KG	2026-10-03 15:08
ORDL-AC-3	OPEN	5000	2026-10-03 14:54	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	58ea4e12	b86353af	a6000000	KG	2026-10-03 14:54
ORDL-AC-2	OPEN	1750	2026-10-03 14:30	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	1902bc3b	38034fe6	50bfcbfa	KG	2026-10-03 14:30
ORDL-AC-1	OPEN	1750	2026-10-03 12:23	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "industry_template_code": "ACTIVATED_CARBON"}	978bddae	38034fe6	a6000000	KG	2026-10-03 12:23

trn.dispatch

Dispatch documents against a customer sales PO.

Primary key: id. Foreign keys: customer_order_id → trn.customer_order(customer_order_id); production_lot_id → trn.production_lot(production_lot_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	

Column	Type	Null	Default
workspace_id	UUID	YES	
customer_order_id	UUID	YES	
production_lot_id	UUID	YES	
finished_goods_lot_no	VARCHAR(128)	YES	
grade	VARCHAR(64)	YES	
bag_count	NUMERIC(18, 4)	YES	
total_weight	NUMERIC(18, 4)	YES	
vehicle_no	VARCHAR(64)	YES	
dispatch_date	DATE	YES	
invoice_ref	VARCHAR(128)	YES	
destination	TEXT	YES	
status	VARCHAR(32)	NO	'PLANNED'::character varying
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NO	CURRENT_TIMESTAMP

No rows in scope (0 total in this filter).

trn.feasibility_check

Feasibility header (UI gate is skipped; ontology rows may still exist).

Primary key: feasibility_check_id. Foreign keys: production_lot_id → trn.production_lot(production_lot_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
feasibility_check_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
check_code	VARCHAR(64)	NO	
production_lot_id	UUID	NO	
overall_status	VARCHAR(32)	NO	'PENDING'::character varying
checked_at	TIMESTAMP	YES	
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (80 of 211 row(s) in GR_CARBONS / catalog scope):

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00211	OK	2026-10-04 19:53	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	e76dc7cf	8f31128b	2026-10-04 19:54	2026-10-04 19:53	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00210	OK	2026-10-04 19:51	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	9ece3350	e0a02f46	2026-10-04 19:54	2026-10-04 19:51	e7000000
FEAS00209	OK	2026-10-04 19:51	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	92858244	8f31128b	2026-10-04 19:53	2026-10-04 19:51	e7000000
FEAS00208	OK	2026-10-04 19:50	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	fa6bf80c	e0a02f46	2026-10-04 19:52	2026-10-04 19:50	e7000000
FEAS00207	OK	2026-10-04 19:50	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	66449a6e	8f31128b	2026-10-04 19:51	2026-10-04 19:50	e7000000
FEAS00206	OK	2026-10-04 19:50	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	838bfc38	e0a02f46	2026-10-04 19:51	2026-10-04 19:50	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00205	OK	2026-10-04 19:49	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	496b50c3	e0a02f46	2026-10-04 19:50	2026-10-04 19:49	e7000000
FEAS00204	OK	2026-10-04 19:49	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	1e5e7429	8f31128b	2026-10-04 19:50	2026-10-04 19:49	e7000000
FEAS00203	OK	2026-10-04 19:49	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	70496816	e0a02f46	2026-10-04 19:50	2026-10-04 19:49	e7000000
FEAS00202	OK	2026-10-04 19:48	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	32bdf9ce	e0a02f46	2026-10-04 19:49	2026-10-04 19:48	e7000000
FEAS00201	OK	2026-10-04 19:47	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	614ad4d0	8f31128b	2026-10-04 19:48	2026-10-04 19:47	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00200	OK	2026-10-04 19:45	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	3c0eca95	8f31128b	2026-10-04 19:47	2026-10-04 19:45	e7000000
FEAS00199	OK	2026-10-04 19:44	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	f7488eed	8f31128b	2026-10-04 19:45	2026-10-04 19:44	e7000000
FEAS00198	OK	2026-10-04 19:37	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	cd43915d	e0a02f46	2026-10-04 19:39	2026-10-04 19:37	e7000000
FEAS00197	OK	2026-10-04 19:36	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	59c2b768	e0a02f46	2026-10-04 19:37	2026-10-04 19:36	e7000000
FEAS00196	OK	2026-10-04 19:35	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	6e340875	e0a02f46	2026-10-04 19:36	2026-10-04 19:35	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00195	OK	2026-10-04 19:30	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	91ca7fe7	e0a02f46	2026-10-04 19:31	2026-10-04 19:30	e7000000
FEAS00194	OK	2026-10-04 19:29	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	ebf72745	e0a02f46	2026-10-04 19:30	2026-10-04 19:29	e7000000
FEAS00193	OK	2026-10-04 19:27	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	6d80be70	e0a02f46	2026-10-04 19:29	2026-10-04 19:27	e7000000
FEAS00192	OK	2026-10-04 17:14	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	53c04fa4	e0a02f46	2026-10-04 17:14	2026-10-04 17:14	e7000000
FEAS00191	OK	2026-10-04 17:14	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	90d60342	e0a02f46	2026-10-04 17:14	2026-10-04 17:14	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00190	OK	2026-10-04 17:14	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	74ebfdf8	e0a02f46	2026-10-04 17:14	2026-10-04 17:14	e7000000
FEAS00189	OK	2026-10-04 17:12	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	3916c08a	e0a02f46	2026-10-04 17:12	2026-10-04 17:12	e7000000
FEAS00188	OK	2026-10-04 17:12	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	b6aa26fb	e0a02f46	2026-10-04 17:12	2026-10-04 17:12	e7000000
FEAS00187	OK	2026-10-04 17:12	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	7af7fc5c	e0a02f46	2026-10-04 17:12	2026-10-04 17:12	e7000000
FEAS00186	OK	2026-10-04 17:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	27e00c0f	e0a02f46	2026-10-04 17:03	2026-10-04 17:03	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00185	OK	2026-10-04 17:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	62db3173	e0a02f46	2026-10-04 17:03	2026-10-04 17:03	e7000000
FEAS00184	OK	2026-10-04 17:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-15, 545.0000 KG)", "status": "OK", "resource_type": "material"},…	5d59c683	36c3c894	2026-10-04 17:03	2026-10-04 17:03	e7000000
FEAS00183	OK	2026-10-04 17:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	ef322490	e0a02f46	2026-10-04 17:03	2026-10-04 17:03	e7000000
FEAS00182	OK	2026-10-04 17:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-15, 545.0000 KG)", "status": "OK", "resource_type": "material"},…	fb7098d6	36c3c894	2026-10-04 17:03	2026-10-04 17:03	e7000000
FEAS00181	OK	2026-10-04 17:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-15, 545.0000 KG)", "status": "OK", "resource_type": "material"},…	6d0e4b19	36c3c894	2026-10-04 17:03	2026-10-04 17:03	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00180	OK	2026-10-04 17:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	a891e52c	e0a02f46	2026-10-04 17:03	2026-10-04 17:03	e7000000
FEAS00179	OK	2026-10-04 17:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	1508f95a	e0a02f46	2026-10-04 17:03	2026-10-04 17:03	e7000000
FEAS00178	OK	2026-10-04 17:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-13 (lot LOT-AC-17, 1500.0000 KG)", "status": "OK", "resource_type": "material"}…	a5b6b9a7	e0a02f46	2026-10-04 17:03	2026-10-04 17:03	e7000000
FEAS00177	OK	2026-10-04 17:01	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	b39a9146	8f31128b	2026-10-04 17:01	2026-10-04 17:01	e7000000
FEAS00176	OK	2026-10-04 17:01	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	6f40cc08	8f31128b	2026-10-04 17:01	2026-10-04 17:01	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00175	OK	2026-10-04 17:01	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	232a10b1	8f31128b	2026-10-04 17:01	2026-10-04 17:01	e7000000
FEAS00174	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	201983d5	8f31128b	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00173	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	663571b8	8f31128b	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00172	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	43d8bc4a	8f31128b	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00171	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	a3d08506	8f31128b	2026-10-04 16:08	2026-10-04 16:08	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00170	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	7ce35ffe	8f31128b	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00169	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	3afc4e0f	8f31128b	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00168	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	7b2e6d12	2b665247	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00167	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	93d2d135	8f31128b	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00166	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	56fa7a89	2b665247	2026-10-04 16:08	2026-10-04 16:08	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00165	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	848285f8	8f31128b	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00164	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	e5729334	2b665247	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00163	OK	2026-10-04 16:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-10 (lot LOT-AC-13, 9000.0000 KG)", "status": "OK", "resource_type": "material"}…	2636b9da	8f31128b	2026-10-04 16:08	2026-10-04 16:08	e7000000
FEAS00162	OK	2026-10-04 15:25	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	bc40afeb	2b665247	2026-10-04 15:25	2026-10-04 15:25	e7000000
FEAS00161	OK	2026-10-04 15:25	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	decbf346	2b665247	2026-10-04 15:25	2026-10-04 15:25	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00160	OK	2026-10-04 15:25	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	7be97617	2b665247	2026-10-04 15:25	2026-10-04 15:25	e7000000
FEAS00159	OK	2026-10-04 15:05	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	78d8a75c	2b665247	2026-10-04 15:06	2026-10-04 15:05	e7000000
FEAS00158	OK	2026-10-04 15:04	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	e575f2c2	2b665247	2026-10-04 15:05	2026-10-04 15:04	e7000000
FEAS00157	OK	2026-10-04 15:03	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	ebe09f7a	2b665247	2026-10-04 15:04	2026-10-04 15:03	e7000000
FEAS00156	OK	2026-10-04 15:02	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	01518ebc	2b665247	2026-10-04 15:03	2026-10-04 15:02	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00155	OK	2026-10-04 14:08	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	e24598e1	2b665247	2026-10-04 14:09	2026-10-04 14:08	e7000000
FEAS00154	OK	2026-10-04 14:07	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	877f21c3	2b665247	2026-10-04 14:08	2026-10-04 14:07	e7000000
FEAS00153	OK	2026-10-04 14:07	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	0da725e4	2b665247	2026-10-04 14:07	2026-10-04 14:07	e7000000
FEAS00152	OK	2026-10-04 13:12	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	b868019d	2b665247	2026-10-04 13:13	2026-10-04 13:12	e7000000
FEAS00151	OK	2026-10-04 13:11	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	09637bd5	2b665247	2026-10-04 13:12	2026-10-04 13:11	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00150	OK	2026-10-04 13:10	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	042937b0	2b665247	2026-10-04 13:11	2026-10-04 13:10	e7000000
FEAS00149	OK	2026-10-04 12:06	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	ecd19ce1	2b665247	2026-10-04 12:06	2026-10-04 12:06	e7000000
FEAS00148	OK	2026-10-04 12:05	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	2b38e653	2b665247	2026-10-04 12:05	2026-10-04 12:05	e7000000
FEAS00147	OK	2026-10-04 12:04	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	2a6e0e4d	2b665247	2026-10-04 12:04	2026-10-04 12:04	e7000000
FEAS00146	OK	2026-10-04 11:58	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	d7f53e2e	2b665247	2026-10-04 11:59	2026-10-04 11:58	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00145	OK	2026-10-04 11:57	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	a5c975a7	2b665247	2026-10-04 11:58	2026-10-04 11:57	e7000000
FEAS00144	OK	2026-10-04 11:56	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	47e1c26f	2b665247	2026-10-04 11:57	2026-10-04 11:56	e7000000
FEAS00143	OK	2026-10-04 10:07	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	d0cb599c	2b665247	2026-10-04 10:08	2026-10-04 10:07	e7000000
FEAS00142	OK	2026-10-04 10:06	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	7374d668	2b665247	2026-10-04 10:07	2026-10-04 10:06	e7000000
FEAS00141	OK	2026-10-04 10:05	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	ff1118f7	2b665247	2026-10-04 10:06	2026-10-04 10:05	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00140	OK	2026-10-04 09:47	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	f206fba8	2b665247	2026-10-04 09:48	2026-10-04 09:47	e7000000
FEAS00139	OK	2026-10-04 09:46	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	11719674	2b665247	2026-10-04 09:47	2026-10-04 09:46	e7000000
FEAS00138	OK	2026-10-04 09:45	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	0ef79fd6	2b665247	2026-10-04 09:46	2026-10-04 09:45	e7000000
FEAS00137	OK	2026-10-04 09:26	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-4 (lot LOT-AC-5, 4999.0000 KG)", "status": "OK", "resource_type": "material"}, …	59741f8f	3ee47ce6	2026-10-04 09:26	2026-10-04 09:26	e7000000
FEAS00136	OK	2026-10-04 09:26	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	d540e063	2b665247	2026-10-04 09:26	2026-10-04 09:26	e7000000

check_code	overall_status	created_at	metadata	feasibility_check_id	production_lot_id	checked_at	updated_at	workspace_id
FEAS00135	OK	2026-10-04 09:26	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-4 (lot LOT-AC-5, 4999.0000 KG)", "status": "OK", "resource_type": "material"}, …	ad783812	3ee47ce6	2026-10-04 09:26	2026-10-04 09:26	e7000000
FEAS00134	OK	2026-10-04 09:26	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	fa7461f3	2b665247	2026-10-04 09:26	2026-10-04 09:26	e7000000
FEAS00133	OK	2026-10-04 09:26	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-4 (lot LOT-AC-5, 4999.0000 KG)", "status": "OK", "resource_type": "material"}, …	7b72e617	3ee47ce6	2026-10-04 09:26	2026-10-04 09:26	e7000000
FEAS00132	OK	2026-10-04 09:26	{"tool": "OK", "checks": [{"detail": null, "status": "OK", "resource_type": "machine"}, {"detail": "Raw material QC passed for order ORD-AC-6 (lot LOT-AC-12, 100.0000 KG)", "status": "OK", "resource_type": "material"}, …	4fea073e	2b665247	2026-10-04 09:26	2026-10-04 09:26	e7000000

131 further row(s) omitted so the PDF stays readable.

trn.feasibility_check_detail

Per-check feasibility lines.

Primary key: feasibility_detail_id. Foreign keys: feasibility_check_id → trn.feasibility_check(feasibility_check_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
feasibility_detail_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
feasibility_check_id	UUID	NO	
check_key	VARCHAR(128)	NO	
check_label	VARCHAR(256)	YES	
status	VARCHAR(32)	NO	'PENDING'::character varying
message	TEXT	YES	

Column	Type	Null	Default
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

No rows in scope (0 total in this filter).

trn.production_lot

Lot created for ontology (LOT-AC-*). Activation / destoning / post-activation state is in metadata.

Primary key: production_lot_id. Foreign keys: production_order_id → trn.production_order(production_order_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
production_lot_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
lot_code	VARCHAR(64)	NO	
production_order_id	UUID	NO	
lot_quantity	NUMERIC(18, 4)	NO	
status	VARCHAR(32)	NO	'CREATED'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (23 of 23 row(s) in GR_CARBONS / catalog scope):

lot_code	status	lot_quantity	created_at	metadata	production_lot_id	production_order_id	updated_at	workspace_id
LOT-AC-23	IN_PROGRESS	2750	2026-10-04 20:52	{"charged_at": "2026-10-04T20:53:16.340939+00:00", "crusher_grade": "Grade 1st Stage", "kiln_loaded_kg": "2750", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "RO_BAG_014", "kiln_feed_events": [{"…	ee5152f0	0f80153a	2026-10-04 20:53	e7000000
LOT-AC-22	IN_PROGRESS	1000	2026-10-04 20:52	{"charged_at": "2026-10-04T20:53:13.080178+00:00", "crusher_grade": "-20 1st Stage", "kiln_loaded_kg": "1000", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "RO_BAG_013", "kiln_feed_events": [{"kg…	283f1784	0f80153a	2026-10-04 20:53	e7000000

lot_code	status	lot_quantity	created_at	metadata	production_lot_id	production_order_id	updated_at	workspace_id
LOT-AC-21	IN_PROGRESS	7500	2026-10-04 19:24	{"charged_at": "2026-10-04T19:57:04.763747+00:00", "kiln_loaded_kg": "1000", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "RO_BAG_001", "kiln_feed_events": [{"kg": 1000.0, "bag_code": "RO_BAG_001…	34e946c5	55e01764	2026-10-04 19:57	e7000000
LOT-AC-20	CREATED	1000	2026-10-04 17:00	{"crusher_grade": "-20 1st Stage - Rotary A", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "BAG-AC-1010", "crusher_outward_bags": [{"bag_code": "BAG-AC-1010", "grade_label": "-20 1st Stage - Rota…	0f0305bb	e0f085f9	2026-10-04 17:00	e7000000
LOT-AC-19	CREATED	1500	2026-10-04 17:00	{"crusher_grade": "Grade 1st stage - Rotary A", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "BAG-AC-1009", "crusher_outward_bags": [{"bag_code": "BAG-AC-1009", "grade_label": "Grade 1st stage - …	a73e42af	e0f085f9	2026-10-04 17:00	e7000000
LOT-AC-18	CREATED	1000	2026-10-04 17:00	{"crusher_grade": "-20 2nd Stage - Rotary B", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "BAG-AC-1008", "crusher_outward_bags": [{"bag_code": "BAG-AC-1008", "grade_label": "-20 2nd Stage - Rota…	c30b46d2	e0f085f9	2026-10-04 17:00	e7000000
LOT-AC-17	IN_PROGRESS	1500	2026-10-04 17:00	{"charged_at": "2026-10-04T17:03:29.093588+00:00", "crusher_grade": "Grade 2nd stage - Rotary B", "kiln_loaded_kg": "1500", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "BAG-AC-1007", "kiln_feed_…	e0a02f46	e0f085f9	2026-10-04 17:03	e7000000

lot_code	status	lot_quantity	created_at	metadata	production_lot_id	production_order_id	updated_at	workspace_id
LOT-AC-16	CREATED	10	2026-10-04 17:00	{"crusher_grade": "Stones", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "BAG-AC-1006", "crusher_outward_bags": [{"bag_code": "BAG-AC-1006", "grade_label": "Stones", "quantity_kg": "10"}], "indus…	e2f59b98	e0f085f9	2026-10-04 17:00	e7000000
LOT-AC-15	IN_PROGRESS	545	2026-10-04 17:00	{"charged_at": "2026-10-04T17:01:43.092097+00:00", "crusher_grade": "Unburnt", "kiln_loaded_kg": "545", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "BAG-AC-1005", "kiln_feed_events": [{"kg": 545…	36c3c894	e0f085f9	2026-10-04 17:01	e7000000
LOT-AC-14	CREATED	999	2026-10-04 16:05	{"workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "BAG-AC-1004", "crusher_outward_bags": [{"bag_code": "BAG-AC-1004", "grade_label": "-20 1st Stage - Rotary A", "quantity_kg": "1000"}, {"bag_code": "…	6714a402	01b3aa33	2026-10-04 16:05	e7000000
LOT-AC-13	IN_PROGRESS	9000	2026-10-04 16:05	{"charged_at": "2026-10-04T16:07:59.800817+00:00", "kiln_loaded_kg": "6000.0", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "crusher_bag_code": "BAG-AC-1004", "kiln_feed_events": [{"kg": 1000.0, "bag_code": "BAG-AC-…	8f31128b	01b3aa33	2026-10-04 19:51	e7000000
LOT-AC-12	IN_PROGRESS	100	2026-10-03 17:36	{"charged_at": "2026-10-03T18:18:08.774276+00:00", "crusher_grade": "Unburnt", "kiln_loaded_kg": "100.0000", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_feed_events": [{"kg": 100.0, "bag_code": null, "lot_cod…	2b665247	edc47c7a	2026-10-05 05:53	e7000000

lot_code	status	lot_quantity	created_at	metadata	production_lot_id	production_order_id	updated_at	workspace_id
LOT-AC-11	IN_PROGRESS	100	2026-10-03 17:36	{"charged_at": "2026-10-04 T02:30:34.289100+00:00", "crusher_grade": "Stones", "kiln_loaded_kg": "100.0000", "workspace_code": "GR_C ARBONS_ACTIVATED_C ARBON", "kiln_feed_events": [{"kg": 100.0, "bag_code": null, "lot_code…	bc18b23d	edc47c7a	2026-10-04 02:30	e7000000
LOT-AC-10	IN_PROGRESS	200	2026-10-03 17:36	{"charged_at": "2026-10-03 T18:30:31.699959+00:00", "crusher_grade": "Grade 2nd stage - Rotary B", "kiln_loaded_kg": "200.0000", "workspace_code": "GR_C ARBONS_ACTIVATED_C ARBON", "kiln_feed_events": [{"kg": 200.0, "bag_co…	57492b12	edc47c7a	2026-10-03 18:38	e7000000
LOT-AC-9	CREATED	200	2026-10-03 17:36	{"crusher_grade": "-20 2nd Stage - Rotary B", "workspace_code": "GR_C ARBONS_ACTIVATED_C ARBON", "industry_template_code": "ACTIVATED_CARBON", "source_inward_order_id": "02d17a82-1fa9-47da-8e64 -0095f18c7cb1"}	d8c94206	edc47c7a	2026-10-03 17:36	e7000000
LOT-AC-8	CREATED	200	2026-10-03 17:36	{"crusher_grade": "Grade 1st stage - Rotary A", "workspace_code": "GR_C ARBONS_ACTIVATED_C ARBON", "industry_template_code": "ACTIVATED_CARBON", "source_inward_order_id": "02d17a82-1fa9-47da-8e64 -0095f18c7cb1"}	da4cd178	edc47c7a	2026-10-03 17:36	e7000000
LOT-AC-7	IN_PROGRESS	200	2026-10-03 17:36	{"charged_at": "2026-10-04 T06:01:12.942778+00:00", "crusher_grade": "-20 1st Stage - Rotary A", "kiln_loaded_kg": "200.0000", "workspace_code": "GR_C ARBONS_ACTIVATED_C ARBON", "kiln_feed_events": [{"kg": 200.0, "bag_code…	1565ddad	edc47c7a	2026-10-04 06:01	e7000000

lot_code	status	lot_quantity	created_at	metadata	production_lot_id	production_order_id	updated_at	workspace_id
LOT-AC-6	IN_PROGRESS	9999	2026-10-03 17:25	{"charged_at": "2026-10-04T02:49:48.787432+00:00", "kiln_loaded_kg": "9999.0000", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_feed_events": [{"kg": 9999.0, "bag_code": null, "lot_code": "LOT-AC-6", "loaded_at…	e58747dc	ebba8b6b	2026-10-04 02:50	e7000000
LOT-AC-5	IN_PROGRESS	4999	2026-10-03 17:19	{"charged_at": "2026-10-04T09:26:00.483220+00:00", "kiln_loaded_kg": "4999.0000", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_feed_events": [{"kg": 4999.0, "bag_code": null, "lot_code": "LOT-AC-5", "loaded_at…	3ee47ce6	af15119e	2026-10-04 09:26	e7000000
LOT-AC-4	IN_PROGRESS	5000	2026-10-03 17:18	{"charged_at": "2026-10-04T06:00:50.610013+00:00", "kiln_loaded_kg": "5000.0000", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_feed_events": [{"kg": 5000.0, "bag_code": null, "lot_code": "LOT-AC-4", "loaded_at…	1acf8ae9	af15119e	2026-10-04 06:00	e7000000
LOT-AC-3	IN_PROGRESS	7777	2026-10-03 15:49	{"charged_at": "2026-10-04T02:46:48.446795+00:00", "kiln_loaded_kg": "7777.0000", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_feed_events": [{"kg": 7777.0, "bag_code": null, "lot_code": "LOT-AC-3", "loaded_at…	7ea895c4	7ece19c6	2026-10-04 02:47	e7000000
LOT-AC-2	IN_PROGRESS	5000	2026-10-03 15:09	{"charged_at": "2026-10-04T02:39:36.781870+00:00", "kiln_loaded_kg": "5000.0000", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_feed_events": [{"kg": 5000.0, "bag_code": null, "lot_code": "LOT-AC-2", "loaded_at…	87279083	002ba9f1	2026-10-04 02:39	e7000000

lot_code	status	lot_quantity	created_at	metadata	production_lot_id	production_order_id	updated_at	workspace_id
LOT-AC-1	IN_PROGRESS	1750	2026-10-03 14:30	{"charged_at": "2026-10-03T18:27:55.621534+00:00", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_live_status": "AVAILABLE", "kiln_resource_code": "RES-KILN-01", "kiln_resource_claimed": false, "industry_templat…	d591ea42	0cb7b89c	2026-10-03 18:28	e7000000

trn.production_order

Internal PO created for ontology when inward is received (not shown on shop floor).

Primary key: production_order_id. Foreign keys: order_line_id → trn.customer_order_line(order_line_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
production_order_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
production_order_code	VARCHAR(64)	NO	
order_line_id	UUID	NO	
planned_quantity	NUMERIC(18, 4)	NO	
status	VARCHAR(32)	NO	'PLANNED'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (10 of 10 row(s) in GR_CARBONS / catalog scope):

production_order_code	status	planned_quantity	created_at	metadata	production_order_id	order_line_id	updated_at	workspace_id
PO-AC-10	PLANNED	3750	2026-10-04 20:52	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	0f80153a	7e4e3d14	2026-10-04 20:52	e7000000
PO-AC-9	PLANNED	7500	2026-10-04 19:24	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	55e01764	f7ac7fa2	2026-10-04 19:24	e7000000

production_order_code	status	planned_quantity	created_at	metadata	production_order_id	order_line_id	updated_at	workspace_id
PO-AC-8	PLANNED	5555	2026-10-04 17:00	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	e0f085f9	ceb69812	2026-10-04 17:00	e7000000
PO-AC-7	PLANNED	9999	2026-10-04 16:05	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	01b3aa33	829fcc44	2026-10-04 16:05	e7000000
PO-AC-6	PLANNED	1000	2026-10-03 17:36	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	edc47c7a	63003eeb	2026-10-03 17:36	e7000000
PO-AC-5	PLANNED	9999	2026-10-03 17:25	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	ebba8b6b	74db2b9b	2026-10-03 17:25	e7000000

production_order_code	status	planned_quantity	created_at	metadata	production_order_id	order_line_id	updated_at	workspace_id
PO-AC-4	PLANNED	9999	2026-10-03 17:18	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	af15119e	b1618bc6	2026-10-03 17:19	e7000000
PO-AC-3	PLANNED	7777	2026-10-03 15:49	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	7ece19c6	85f957c8	2026-10-03 15:49	e7000000
PO-AC-2	PLANNED	5000	2026-10-03 15:09	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	002ba9f1	3a1011ee	2026-10-03 15:09	e7000000
PO-AC-1	PLANNED	1750	2026-10-03 14:30	{"flow": "RELEASED_TO_PRODUCTION", "routing_id": "8b1ea279-7bde-4675-8ea7-0adc48d67fa8", "routing_code": "RT-AC-2", "raw_product_id": "a6000000-0000-4000-8000-000000000311", "workspace_code": "GR_CARBONS_ACTIVATED_CARBO…	0cb7b89c	1902bc3b	2026-10-03 14:30	e7000000

trn.sensor_reading

Operational / catalog table.

Primary key: id. Foreign keys: device_id → mst.sensor_device(id).

Column	Type	Null	Default
id	BIGINT	NO	nextval('trn.sensor_reading_id_seq'::re…
device_id	BIGINT	NO	

Column	Type	Null	Default
reading_timestamp	TIMESTAMP	NO	
source	VARCHAR(50)	NO	'SMART_MODBUS_SIMULATOR'::character var…
payload_format	VARCHAR(50)	NO	'SMART_MODBUS_JSON'::character varying
protocol	VARCHAR(50)	NO	'MODBUS_TCP'::character varying
quality_status	VARCHAR(50)	NO	'GOOD'::character varying
raw_payload	JSONB	YES	
created_at	TIMESTAMP	NO	now()

Data (80 of 753 row(s) in GR_CARBONS / catalog scope):

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 16:05	753	1	2026-08-09 16:05	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 839.3, "quality": "GOOD", "slave_id": 1, "raw_value": "839.3045398203445", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 16:04	752	1	2026-08-09 16:04	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 758.5, "quality": "GOOD", "slave_id": 1, "raw_value": "758.4705367155223", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 16:03	751	1	2026-08-09 16:03	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 760.6, "quality": "GOOD", "slave_id": 1, "raw_value": "760.5550754259432", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 16:01	750	1	2026-08-09 16:01	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 777.4, "quality": "GOOD", "slave_id": 1, "raw_value": "777.4202999184929", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 16:00	749	1	2026-08-09 16:00	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 850.2, "quality": "GOOD", "slave_id": 1, "raw_value": "850.2128636614804", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:59	748	1	2026-08-09 15:59	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 936.8, "quality": "GOOD", "slave_id": 1, "raw_value": "936.801453089975", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 15:58	747	1	2026-08-09 15:58	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 834.0, "quality": "GOOD", "slave_id": 1, "raw_value": "833.9797300087412", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:54	746	1	2026-08-09 15:54	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 817.6, "quality": "GOOD", "slave_id": 1, "raw_value": "817.5877299641324", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:53	745	1	2026-08-09 15:53	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 913.7, "quality": "GOOD", "slave_id": 1, "raw_value": "913.72027860818", "normal_max": 950.0, "normal_min": 750.0, "fun…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 15:52	744	1	2026-08-09 15:52	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 897.0, "quality": "GOOD", "slave_id": 1, "raw_value": "897.0235343862187", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:51	743	1	2026-08-09 15:51	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 869.1, "quality": "GOOD", "slave_id": 1, "raw_value": "869.0706595700453", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:49	742	1	2026-08-09 15:49	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 755.4, "quality": "GOOD", "slave_id": 1, "raw_value": "755.4330803296891", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:48	741	1	2026-08-09 15:48	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 814.6, "quality": "GOOD", "slave_id": 1, "raw_value": "814.648027300102", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 15:47	740	1	2026-08-09 15:47	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 823.2, "quality": "GOOD", "slave_id": 1, "raw_value": "823.198655345209", "normal_max": 950.0, "normal_min": 750.0, "fu…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 15:46	739	1	2026-08-09 15:46	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 947.7, "quality": "GOOD", "slave_id": 1, "raw_value": "947.7476656300289", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:45	738	1	2026-08-09 15:45	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 884.1, "quality": "GOOD", "slave_id": 1, "raw_value": "884.0951329941829", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:44	737	1	2026-08-09 15:44	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 764.2, "quality": "GOOD", "slave_id": 1, "raw_value": "764.239632564615", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 15:43	736	1	2026-08-09 15:43	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 761.7, "quality": "GOOD", "slave_id": 1, "raw_value": "761.6683431704796", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:42	735	1	2026-08-09 15:42	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 939.4, "quality": "GOOD", "slave_id": 1, "raw_value": "939.3526519113876", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 15:41	734	1	2026-08-09 15:41	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 871.0, "quality": "GOOD", "slave_id": 1, "raw_value": "871.0178758656144", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:40	733	1	2026-08-09 15:40	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 881.5, "quality": "GOOD", "slave_id": 1, "raw_value": "881.4876430697938", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:39	732	1	2026-08-09 15:39	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 820.8, "quality": "GOOD", "slave_id": 1, "raw_value": "820.8493266634065", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:38	731	1	2026-08-09 15:38	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 869.3, "quality": "GOOD", "slave_id": 1, "raw_value": "869.3411514235264", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:37	730	1	2026-08-09 15:37	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 790.7, "quality": "GOOD", "slave_id": 1, "raw_value": "790.7004360471806", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 15:36	729	1	2026-08-09 15:36	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 785.4, "quality": "GOOD", "slave_id": 1, "raw_value": "785.4162392318323", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:35	728	1	2026-08-09 15:35	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 825.8, "quality": "GOOD", "slave_id": 1, "raw_value": "825.7944233297914", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:34	727	1	2026-08-09 15:34	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 758.0, "quality": "GOOD", "slave_id": 1, "raw_value": "758.0382925580245", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:33	726	1	2026-08-09 15:33	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 885.4, "quality": "GOOD", "slave_id": 1, "raw_value": "885.446522812181", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 15:32	725	1	2026-08-09 15:32	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 940.2, "quality": "GOOD", "slave_id": 1, "raw_value": "940.1683114271598", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 15:31	724	1	2026-08-09 15:31	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 946.8, "quality": "GOOD", "slave_id": 1, "raw_value": "946.7598629271864", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:30	723	1	2026-08-09 15:30	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 858.0, "quality": "GOOD", "slave_id": 1, "raw_value": "858.0034649022067", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:29	722	1	2026-08-09 15:29	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 758.9, "quality": "GOOD", "slave_id": 1, "raw_value": "758.9053383769752", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:28	721	1	2026-08-09 15:28	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 821.7, "quality": "GOOD", "slave_id": 1, "raw_value": "821.7136023408326", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:27	720	1	2026-08-09 15:27	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 782.5, "quality": "GOOD", "slave_id": 1, "raw_value": "782.4744415656955", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 15:26	719	1	2026-08-09 15:26	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 830.6, "quality": "GOOD", "slave_id": 1, "raw_value": "830.5926952948226", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:25	718	1	2026-08-09 15:25	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 800.7, "quality": "GOOD", "slave_id": 1, "raw_value": "800.6717213810505", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:24	717	1	2026-08-09 15:24	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 765.2, "quality": "GOOD", "slave_id": 1, "raw_value": "765.1516083142337", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 15:23	716	1	2026-08-09 15:23	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 793.8, "quality": "GOOD", "slave_id": 1, "raw_value": "793.7733811925957", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 11:11	715	1	2026-08-09 11:11	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 891.1, "quality": "GOOD", "slave_id": 1, "raw_value": "891.1206356671044", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 11:10	714	1	2026-08-09 11:10	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 786.8, "quality": "GOOD", "slave_id": 1, "raw_value": "786.7837323711044", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 11:09	713	1	2026-08-09 11:09	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 833.7, "quality": "GOOD", "slave_id": 1, "raw_value": "833.7163356036417", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 11:08	712	1	2026-08-09 11:08	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 776.1, "quality": "GOOD", "slave_id": 1, "raw_value": "776.0819692458276", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 11:07	711	1	2026-08-09 11:07	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 767.3, "quality": "GOOD", "slave_id": 1, "raw_value": "767.3242598005419", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 11:06	710	1	2026-08-09 11:06	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 818.1, "quality": "GOOD", "slave_id": 1, "raw_value": "818.1190284621463", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 11:05	709	1	2026-08-09 11:05	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 762.3, "quality": "GOOD", "slave_id": 1, "raw_value": "762.2724708126282", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 11:04	708	1	2026-08-09 11:04	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 944.3, "quality": "GOOD", "slave_id": 1, "raw_value": "944.344466299746", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 11:02	707	1	2026-08-09 11:02	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 929.6, "quality": "GOOD", "slave_id": 1, "raw_value": "929.647893107229", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 11:01	706	1	2026-08-09 11:01	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 940.3, "quality": "GOOD", "slave_id": 1, "raw_value": "940.3440779651964", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 11:00	705	1	2026-08-09 11:00	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 873.3, "quality": "GOOD", "slave_id": 1, "raw_value": "873.2793429433497", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 10:59	704	1	2026-08-09 10:59	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 949.3, "quality": "GOOD", "slave_id": 1, "raw_value": "949.2593441814084", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:58	703	1	2026-08-09 10:58	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 876.3, "quality": "GOOD", "slave_id": 1, "raw_value": "876.2534218207123", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:57	702	1	2026-08-09 10:57	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 800.9, "quality": "GOOD", "slave_id": 1, "raw_value": "800.8896575895872", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:56	701	1	2026-08-09 10:56	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 873.5, "quality": "GOOD", "slave_id": 1, "raw_value": "873.4602366566185", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:55	700	1	2026-08-09 10:55	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 944.5, "quality": "GOOD", "slave_id": 1, "raw_value": "944.4567983307389", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 10:54	699	1	2026-08-09 10:54	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 927.4, "quality": "GOOD", "slave_id": 1, "raw_value": "927.4355650189837", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:53	698	1	2026-08-09 10:53	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 807.1, "quality": "GOOD", "slave_id": 1, "raw_value": "807.1328557698556", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:52	697	1	2026-08-09 10:52	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 914.1, "quality": "GOOD", "slave_id": 1, "raw_value": "914.077564301031", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 10:51	696	1	2026-08-09 10:51	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 751.1, "quality": "GOOD", "slave_id": 1, "raw_value": "751.0633572560483", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:50	695	1	2026-08-09 10:50	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 760.6, "quality": "GOOD", "slave_id": 1, "raw_value": "760.5566381650133", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 10:49	694	1	2026-08-09 10:49	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 842.8, "quality": "GOOD", "slave_id": 1, "raw_value": "842.7702521470426", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:48	693	1	2026-08-09 10:48	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 914.1, "quality": "GOOD", "slave_id": 1, "raw_value": "914.1328021434765", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:47	692	1	2026-08-09 10:47	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 779.3, "quality": "GOOD", "slave_id": 1, "raw_value": "779.2564259405588", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:46	691	1	2026-08-09 10:46	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 872.4, "quality": "GOOD", "slave_id": 1, "raw_value": "872.4003391424361", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:45	690	1	2026-08-09 10:45	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 772.2, "quality": "GOOD", "slave_id": 1, "raw_value": "772.1959716992708", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 10:44	689	1	2026-08-09 10:44	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 801.7, "quality": "GOOD", "slave_id": 1, "raw_value": "801.7206359915369", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:43	688	1	2026-08-09 10:43	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 942.3, "quality": "GOOD", "slave_id": 1, "raw_value": "942.3454043786575", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:42	687	1	2026-08-09 10:42	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 855.2, "quality": "GOOD", "slave_id": 1, "raw_value": "855.2115907672026", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:41	686	1	2026-08-09 10:41	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 829.8, "quality": "GOOD", "slave_id": 1, "raw_value": "829.839497335262", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 10:40	685	1	2026-08-09 10:40	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 844.9, "quality": "GOOD", "slave_id": 1, "raw_value": "844.898122197416", "normal_max": 950.0, "normal_min": 750.0, "fu…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 10:39	684	1	2026-08-09 10:39	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 873.2, "quality": "GOOD", "slave_id": 1, "raw_value": "873.2075474776938", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:38	683	1	2026-08-09 10:38	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 923.4, "quality": "GOOD", "slave_id": 1, "raw_value": "923.353491233847", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 10:37	682	1	2026-08-09 10:37	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 805.2, "quality": "GOOD", "slave_id": 1, "raw_value": "805.1769808913338", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:36	681	1	2026-08-09 10:36	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 766.6, "quality": "GOOD", "slave_id": 1, "raw_value": "766.6049927065403", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:35	680	1	2026-08-09 10:35	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 783.3, "quality": "GOOD", "slave_id": 1, "raw_value": "783.2539123834613", "normal_max": 950.0, "normal_min": 750.0, "f…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 10:34	679	1	2026-08-09 10:34	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 813.1, "quality": "GOOD", "slave_id": 1, "raw_value": "813.0707618891233", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:33	678	1	2026-08-09 10:33	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 781.3, "quality": "GOOD", "slave_id": 1, "raw_value": "781.3101694340288", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:32	677	1	2026-08-09 10:32	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 762.4, "quality": "GOOD", "slave_id": 1, "raw_value": "762.372844142972", "normal_max": 950.0, "normal_min": 750.0, "fu…
2026-08-09 10:31	676	1	2026-08-09 10:31	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 876.5, "quality": "GOOD", "slave_id": 1, "raw_value": "876.4584144652197", "normal_max": 950.0, "normal_min": 750.0, "f…
2026-08-09 10:30	675	1	2026-08-09 10:30	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 861.7, "quality": "GOOD", "slave_id": 1, "raw_value": "861.685604293855", "normal_max": 950.0, "normal_min": 750.0, "fu…

created_at	id	device_id	reading_timestamp	source	payload_format	protocol	quality_status	raw_payload
2026-08-09 10:28	674	1	2026-08-09 10:28	SMART_MODBUS_SIMULATOR	SMART_MODBUS_JSON	MODBUS_TCP	GOOD	{"source": "SMART_MODBUS_SIMULATOR", "protocol": "MODBUS_TCP", "parameters": [{"uom": "C", "value": 765.8, "quality": "GOOD", "slave_id": 1, "raw_value": "765.8140259509499", "normal_max": 950.0, "normal_min": 750.0, "f…

673 further row(s) omitted so the PDF stays readable.

trn.sensor_reading_value

Operational / catalog table.

Primary key: id. Foreign keys: parameter_id → mst.sensor_parameter(id); sensor_reading_id → trn.sensor_reading(id).

Column	Type	Null	Default
id	BIGINT	NO	nextval('trn.sensor_reading_value_id_se…
sensor_reading_id	BIGINT	NO	
parameter_id	BIGINT	NO	
parameter_code	VARCHAR(100)	NO	
value_numeric	NUMERIC	YES	
value_text	VARCHAR(200)	YES	
uom	VARCHAR(50)	YES	
quality	VARCHAR(50)	NO	'GOOD'::character varying
slave_id	INTEGER	YES	
function_code	INTEGER	YES	
register_address	INTEGER	YES	
raw_value	VARCHAR(200)	YES	
created_at	TIMESTAMP	NO	now()

Data (80 of 5271 row(s) in GR_CARBONS / catalog scope):

created_at	id	sensor_reading_id	parameter_id	parameter_code	value_numeric	value_text	uom	quality	slave_id	function_code	register_address
2026-08-09 16:05	5271	753	7	RESIDENCE_TIME	38.6		hr	GOOD	1	3	40013
2026-08-09 16:05	5270	753	6	STEAM_FLOW	168.8		kg/hr	GOOD	1	3	40011
2026-08-09 16:05	5269	753	5	PRESSURE	1.4		bar	GOOD	1	3	40009
2026-08-09 16:05	5268	753	4	TEMP_ZONE_4	787.3		C	GOOD	1	3	40007
2026-08-09 16:05	5267	753	3	TEMP_ZONE_3	862		C	GOOD	1	3	40005
2026-08-09 16:05	5266	753	2	TEMP_ZONE_2	851		C	GOOD	1	3	40003
2026-08-09 16:05	5265	753	1	TEMP_ZONE_1	839.3		C	GOOD	1	3	40001
2026-08-09 16:04	5264	752	7	RESIDENCE_TIME	36.7		hr	GOOD	1	3	40013
2026-08-09 16:04	5263	752	6	STEAM_FLOW	202.3		kg/hr	GOOD	1	3	40011
2026-08-09 16:04	5262	752	5	PRESSURE	1		bar	GOOD	1	3	40009
2026-08-09 16:04	5261	752	4	TEMP_ZONE_4	937.4		C	GOOD	1	3	40007
2026-08-09 16:04	5260	752	3	TEMP_ZONE_3	879.6		C	GOOD	1	3	40005

created_at	id	sensor_reading_id	parameter_id	parameter_code	value_numeric	value_text	uom	quality	slave_id	function_code	register_address
2026-08-09 16:04	5259	752	2	TEMP_ZONE_2	923		C	GOOD	1	3	40003
2026-08-09 16:04	5258	752	1	TEMP_ZONE_1	758.5		C	GOOD	1	3	40001
2026-08-09 16:03	5257	751	7	RESIDENCE_TIME	38.4		hr	GOOD	1	3	40013
2026-08-09 16:03	5256	751	6	STEAM_FLOW	285.5		kg/hr	GOOD	1	3	40011
2026-08-09 16:03	5255	751	5	PRESSURE	0.8		bar	GOOD	1	3	40009
2026-08-09 16:03	5254	751	4	TEMP_ZONE_4	817.6		C	GOOD	1	3	40007
2026-08-09 16:03	5253	751	3	TEMP_ZONE_3	824.4		C	GOOD	1	3	40005
2026-08-09 16:03	5252	751	2	TEMP_ZONE_2	797.3		C	GOOD	1	3	40003
2026-08-09 16:03	5251	751	1	TEMP_ZONE_1	760.6		C	GOOD	1	3	40001
2026-08-09 16:01	5250	750	7	RESIDENCE_TIME	35.7		hr	GOOD	1	3	40013
2026-08-09 16:01	5249	750	6	STEAM_FLOW	135		kg/hr	GOOD	1	3	40011
2026-08-09 16:01	5248	750	5	PRESSURE	1.5		bar	GOOD	1	3	40009
2026-08-09 16:01	5247	750	4	TEMP_ZONE_4	777.9		C	GOOD	1	3	40007
2026-08-09 16:01	5246	750	3	TEMP_ZONE_3	852.4		C	GOOD	1	3	40005
2026-08-09 16:01	5245	750	2	TEMP_ZONE_2	876.1		C	GOOD	1	3	40003
2026-08-09 16:01	5244	750	1	TEMP_ZONE_1	777.4		C	GOOD	1	3	40001
2026-08-09 16:00	5243	749	7	RESIDENCE_TIME	38.3		hr	GOOD	1	3	40013
2026-08-09 16:00	5242	749	6	STEAM_FLOW	123.4		kg/hr	GOOD	1	3	40011
2026-08-09 16:00	5241	749	5	PRESSURE	1.2		bar	GOOD	1	3	40009
2026-08-09 16:00	5240	749	4	TEMP_ZONE_4	757.9		C	GOOD	1	3	40007
2026-08-09 16:00	5239	749	3	TEMP_ZONE_3	889.9		C	GOOD	1	3	40005
2026-08-09 16:00	5238	749	2	TEMP_ZONE_2	870.8		C	GOOD	1	3	40003
2026-08-09 16:00	5237	749	1	TEMP_ZONE_1	850.2		C	GOOD	1	3	40001
2026-08-09 15:59	5236	748	7	RESIDENCE_TIME	24.5		hr	GOOD	1	3	40013
2026-08-09 15:59	5235	748	6	STEAM_FLOW	251.2		kg/hr	GOOD	1	3	40011
2026-08-09 15:59	5234	748	5	PRESSURE	1		bar	GOOD	1	3	40009
2026-08-09 15:59	5233	748	4	TEMP_ZONE_4	822.3		C	GOOD	1	3	40007
2026-08-09 15:59	5232	748	3	TEMP_ZONE_3	802.6		C	GOOD	1	3	40005
2026-08-09 15:59	5231	748	2	TEMP_ZONE_2	856.1		C	GOOD	1	3	40003
2026-08-09 15:59	5230	748	1	TEMP_ZONE_1	936.8		C	GOOD	1	3	40001
2026-08-09 15:58	5229	747	7	RESIDENCE_TIME	30.5		hr	GOOD	1	3	40013
2026-08-09 15:58	5228	747	6	STEAM_FLOW	167.7		kg/hr	GOOD	1	3	40011
2026-08-09 15:58	5227	747	5	PRESSURE	1.3		bar	GOOD	1	3	40009
2026-08-09 15:58	5226	747	4	TEMP_ZONE_4	820.2		C	GOOD	1	3	40007
2026-08-09 15:58	5225	747	3	TEMP_ZONE_3	797		C	GOOD	1	3	40005
2026-08-09 15:58	5224	747	2	TEMP_ZONE_2	928.9		C	GOOD	1	3	40003
2026-08-09 15:58	5223	747	1	TEMP_ZONE_1	834		C	GOOD	1	3	40001
2026-08-09 15:54	5222	746	7	RESIDENCE_TIME	25.9		hr	GOOD	1	3	40013
2026-08-09 15:54	5221	746	6	STEAM_FLOW	253.9		kg/hr	GOOD	1	3	40011
2026-08-09 15:54	5220	746	5	PRESSURE	0.9		bar	GOOD	1	3	40009
2026-08-09 15:54	5219	746	4	TEMP_ZONE_4	893.5		C	GOOD	1	3	40007

created_at	id	sensor_reading_id	parameter_id	parameter_code	value_numeric	value_text	uom	quality	slave_id	function_code	register_address
2026-08-09 15:54	5218	746	3	TEMP_ZONE_3	796.3		C	GOOD	1	3	40005
2026-08-09 15:54	5217	746	2	TEMP_ZONE_2	857.9		C	GOOD	1	3	40003
2026-08-09 15:54	5216	746	1	TEMP_ZONE_1	817.6		C	GOOD	1	3	40001
2026-08-09 15:53	5215	745	7	RESIDENCE_TIME	35.2		hr	GOOD	1	3	40013
2026-08-09 15:53	5214	745	6	STEAM_FLOW	222.1		kg/hr	GOOD	1	3	40011
2026-08-09 15:53	5213	745	5	PRESSURE	0.9		bar	GOOD	1	3	40009
2026-08-09 15:53	5212	745	4	TEMP_ZONE_4	941.5		C	GOOD	1	3	40007
2026-08-09 15:53	5211	745	3	TEMP_ZONE_3	909.3		C	GOOD	1	3	40005
2026-08-09 15:53	5210	745	2	TEMP_ZONE_2	940.8		C	GOOD	1	3	40003
2026-08-09 15:53	5209	745	1	TEMP_ZONE_1	913.7		C	GOOD	1	3	40001
2026-08-09 15:52	5208	744	7	RESIDENCE_TIME	32		hr	GOOD	1	3	40013
2026-08-09 15:52	5207	744	6	STEAM_FLOW	139.5		kg/hr	GOOD	1	3	40011
2026-08-09 15:52	5206	744	5	PRESSURE	1.1		bar	GOOD	1	3	40009
2026-08-09 15:52	5205	744	4	TEMP_ZONE_4	923.6		C	GOOD	1	3	40007
2026-08-09 15:52	5204	744	3	TEMP_ZONE_3	798.6		C	GOOD	1	3	40005
2026-08-09 15:52	5203	744	2	TEMP_ZONE_2	759.8		C	GOOD	1	3	40003
2026-08-09 15:52	5202	744	1	TEMP_ZONE_1	897		C	GOOD	1	3	40001
2026-08-09 15:51	5201	743	7	RESIDENCE_TIME	22.2		hr	GOOD	1	3	40013
2026-08-09 15:51	5200	743	6	STEAM_FLOW	296.8		kg/hr	GOOD	1	3	40011
2026-08-09 15:51	5199	743	5	PRESSURE	1.3		bar	GOOD	1	3	40009
2026-08-09 15:51	5198	743	4	TEMP_ZONE_4	794.5		C	GOOD	1	3	40007
2026-08-09 15:51	5197	743	3	TEMP_ZONE_3	909.2		C	GOOD	1	3	40005
2026-08-09 15:51	5196	743	2	TEMP_ZONE_2	855.7		C	GOOD	1	3	40003
2026-08-09 15:51	5195	743	1	TEMP_ZONE_1	869.1		C	GOOD	1	3	40001
2026-08-09 15:49	5194	742	7	RESIDENCE_TIME	28.4		hr	GOOD	1	3	40013
2026-08-09 15:49	5193	742	6	STEAM_FLOW	182		kg/hr	GOOD	1	3	40011
2026-08-09 15:49	5192	742	5	PRESSURE	1.1		bar	GOOD	1	3	40009

5191 further row(s) omitted so the PDF stays readable.

exec.operation_execution

Shop-floor operation run for a lot + routing step.

Primary key: execution_id. Foreign keys: equipment_id → mst.equipment(equipment_id); production_lot_id → trn.production_lot(production_lot_id); routing_operation_id → mst.routing_operation(routing_operation_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
execution_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
execution_code	VARCHAR(64)	NO	
production_lot_id	UUID	NO	
routing_operation_id	UUID	NO	
equipment_id	UUID	YES	
status	VARCHAR(32)	NO	'NOT_STARTED'::character varying

Column	Type	Null	Default
started_at	TIMESTAMP	YES	
completed_at	TIMESTAMP	YES	
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (80 of 185 row(s) in GR_CARBONS / catalog scope):

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
COMPLETED	2026-10-04 20:53	{"process_step": "KILN_FEED", "feed_recorded": true, "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_resource_code": "RES-KILN-0002", "industry_template_code": "ACTIVATED_CARBON"}	47ff5cf1	EXEC-AC-172	ee5152f0	64d837bc				2026-10-04 20:53
COMPLETED	2026-10-04 20:53	{"process_step": "KILN_FEED", "feed_recorded": true, "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_resource_code": "RES-KILN-0002", "industry_template_code": "ACTIVATED_CARBON"}	bc466c67	EXEC-AC-160	283f1784	64d837bc				2026-10-04 20:53
COMPLETED	2026-10-04 19:57	{"process_step": "KILN_FEED", "feed_recorded": true, "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "kiln_resource_code": "RES-KILN-0002", "industry_template_code": "ACTIVATED_CARBON"}	17765c6e	EXEC-AC-159	34e946c5	64d837bc				2026-10-04 19:57
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.920278+00:00", "is_mandatory": true, "process_step": "DESTONING", "schedule_code": "WO00172", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	40134507	WO00172	36c3c894	a25971a3				2026-10-04 17:03

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.922561+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00177", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	5f1d4817	WO00177	36c3c894	0c7bf68a				2026-10-04 17:03
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.920686+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00173", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	0abf0c5f	WO00173	36c3c894	86967ce3				2026-10-04 17:03
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.923041+00:00", "is_mandatory": true, "process_step": "GRADE_SEPARATION", "schedule_code": "WO00178", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON",…	31a04471	WO00178	36c3c894	63f60409				2026-10-04 17:03
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.923534+00:00", "is_mandatory": true, "process_step": "ACTIVATION", "schedule_code": "WO00179", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sche…	c1c00b17	WO00179	36c3c894	7b6628b6				2026-10-04 17:03

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.923994+00:00", "is_mandatory": true, "process_step": "QC", "schedule_code": "WO00180", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_sta…	fded80de	WO00180	36c3c894	718a96c8				2026-10-04 17:03
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.921610+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00175", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	eb4c6a77	WO00175	36c3c894	067fdcfe				2026-10-04 17:03
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.919731+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00171", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	e9bf9eb9	WO00171	36c3c894	cab75d16				2026-10-04 17:03
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.921143+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00174", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	04712552	WO00174	36c3c894	ab8f805e				2026-10-04 17:03

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.922031+00:00", "is_mandatory": true, "process_step": "DRYING", "schedule_code": "WO00176", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule…	adb90e2a	WO00176	36c3c894	9cc0842f				2026-10-04 17:03
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:31.919111+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00170", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	f18e2e7a	WO00170	36c3c894	caf7b3c2				2026-10-04 17:03
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.442008+00:00", "is_mandatory": true, "process_step": "ACTIVATION", "schedule_code": "WO00168", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sche…	d6c02bea	WO00168	e0a02f46	7b6628b6				2026-10-04 19:51
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.437657+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00159", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	fd95eb30	WO00159	e0a02f46	caf7b3c2				2026-10-04 19:51

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.438180+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00160", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	52a4bcd4	WO00160	e0a02f46	cab75d16				2026-10-04 19:51
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.441025+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00166", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	c4fc8f35	WO00166	e0a02f46	0c7bf68a				2026-10-04 19:51
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.439966+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00164", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	efc1c30d	WO00164	e0a02f46	067fdcfe				2026-10-04 19:51
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.441577+00:00", "is_mandatory": true, "process_step": "GRADE_SEPARATION", "schedule_code": "WO00167", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON",…	8404eacb	WO00167	e0a02f46	63f60409				2026-10-04 19:51

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.439589+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00163", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	a42933b3	WO00163	e0a02f46	ab8f805e				2026-10-04 19:51
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.442566+00:00", "is_mandatory": true, "process_step": "QC", "schedule_code": "WO00169", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_sta…	50726d95	WO00169	e0a02f46	718a96c8				2026-10-04 19:51
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.438734+00:00", "is_mandatory": true, "process_step": "DESTONING", "schedule_code": "WO00161", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	fe32c6be	WO00161	e0a02f46	a25971a3				2026-10-04 19:51
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.439169+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00162", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	03f821da	WO00162	e0a02f46	86967ce3				2026-10-04 19:51

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 17:03	{"source": "ac_routing_generate", "released_at": "2026-10-04T17:03:29.440438+00:00", "is_mandatory": true, "process_step": "DRYING", "schedule_code": "WO00165", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule…	48c94ad5	WO00165	e0a02f46	9cc0842f				2026-10-04 19:51
COMPLETED	2026-10-04 17:03	{"source": "ac_routing_generate", "is_mandatory": true, "process_step": "ACTIVATION", "feed_recorded": true, "schedule_code": "EXEC-AC-158", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_status": "COMPLETED…	6788e8ac	EXEC-AC-158	e0a02f46	64d837bc				2026-10-04 19:51
COMPLETED	2026-10-04 17:01	{"source": "ac_routing_generate", "is_mandatory": true, "process_step": "ACTIVATION", "feed_recorded": true, "schedule_code": "EXEC-AC-135", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_status": "COMPLETED…	045a35e0	EXEC-AC-135	36c3c894	64d837bc				2026-10-04 17:03
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.383753+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00146", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	46ef605e	WO00146	8f31128b	caf7b3c2				2026-10-04 19:54

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.958826+00:00", "is_mandatory": true, "process_step": "DRYING", "schedule_code": "WO00152", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule…	b7d627b9	WO00152	8f31128b	9cc0842f				2026-10-04 19:54
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.977371+00:00", "is_mandatory": true, "process_step": "GRADE_SEPARATION", "schedule_code": "WO00154", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON",…	7bb9ede9	WO00154	8f31128b	63f60409				2026-10-04 19:54
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.436242+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00149", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	d2ffe5e9	WO00149	8f31128b	86967ce3				2026-10-04 19:54
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.981332+00:00", "is_mandatory": true, "process_step": "ACTIVATION", "schedule_code": "WO00155", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sche…	543e7782	WO00155	8f31128b	7b6628b6				2026-10-04 19:54

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.965028+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00153", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	6725ae27	WO00153	8f31128b	0c7bf68a				2026-10-04 19:54
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.476076+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00150", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	8c4c1bc8	WO00150	8f31128b	ab8f805e				2026-10-04 19:54
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.399086+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00147", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	83a9e476	WO00147	8f31128b	cab75d16				2026-10-04 19:54
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.426810+00:00", "is_mandatory": true, "process_step": "DESTONING", "schedule_code": "WO00148", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	c668213a	WO00148	8f31128b	a25971a3				2026-10-04 19:54

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.994187+00:00", "is_mandatory": true, "process_step": "QC", "schedule_code": "WO00156", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_sta…	5f019a38	WO00156	8f31128b	718a96c8				2026-10-04 19:54
NOT_STARTED	2026-10-04 16:08	{"source": "ac_routing_generate", "released_at": "2026-10-04T16:08:01.485465+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00151", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	3f032495	WO00151	8f31128b	067fdcfe				2026-10-04 19:54
COMPLETED	2026-10-04 16:07	{"source": "ac_routing_generate", "is_mandatory": true, "process_step": "ACTIVATION", "feed_recorded": true, "schedule_code": "EXEC-AC-134", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_status": "COMPLETED…	cda121f3	EXEC-AC-134	8f31128b	64d837bc				2026-10-04 19:54
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.813437+00:00", "is_mandatory": true, "process_step": "GRADE_SEPARATION", "schedule_code": "WO00142", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON",…	951af5ee	WO00142	3ee47ce6	63f60409				2026-10-04 09:26

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.812981+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00141", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	d340e0b2	WO00141	3ee47ce6	0c7bf68a				2026-10-04 09:26
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.810276+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00135", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	be036666	WO00135	3ee47ce6	cab75d16				2026-10-04 09:26
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.809757+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00134", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	fa78c380	WO00134	3ee47ce6	caf7b3c2				2026-10-04 09:26
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.811697+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00138", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	2ec57bb2	WO00138	3ee47ce6	ab8f805e				2026-10-04 09:26

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.814529+00:00", "is_mandatory": true, "process_step": "QC", "schedule_code": "WO00144", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_sta…	41353d7a	WO00144	3ee47ce6	718a96c8				2026-10-04 09:26
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.810843+00:00", "is_mandatory": true, "process_step": "DESTONING", "schedule_code": "WO00136", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	888609b6	WO00136	3ee47ce6	a25971a3				2026-10-04 09:26
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.812582+00:00", "is_mandatory": true, "process_step": "DRYING", "schedule_code": "WO00140", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule…	66fd010a	WO00140	3ee47ce6	9cc0842f				2026-10-04 09:26
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.812177+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00139", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	0cca8776	WO00139	3ee47ce6	067fdcfe				2026-10-04 09:26

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.811289+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00137", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	0087f8b5	WO00137	3ee47ce6	86967ce3				2026-10-04 09:26
NOT_STARTED	2026-10-04 09:26	{"source": "ac_routing_generate", "released_at": "2026-10-04T09:26:00.813931+00:00", "is_mandatory": true, "process_step": "ACTIVATION", "schedule_code": "WO00143", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sche…	74d05302	WO00143	3ee47ce6	7b6628b6				2026-10-04 09:26
COMPLETED	2026-10-04 09:26	{"source": "ac_routing_generate", "is_mandatory": true, "process_step": "ACTIVATION", "feed_recorded": true, "schedule_code": "EXEC-AC-133", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_status": "COMPLETED…	ccc7013c	EXEC-AC-133	3ee47ce6	64d837bc				2026-10-04 09:26
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.224112+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00125", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	44cfcab5	WO00125	1565ddad	86967ce3				2026-10-04 06:01

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.223616+00:00", "is_mandatory": true, "process_step": "DESTONING", "schedule_code": "WO00124", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	c6a3adfc	WO00124	1565ddad	a25971a3				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.225382+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00128", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	4e9b9ee8	WO00128	1565ddad	0c7bf68a				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.224569+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00126", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	94cf0e06	WO00126	1565ddad	067fdcfe				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.224985+00:00", "is_mandatory": true, "process_step": "DRYING", "schedule_code": "WO00127", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule…	11f88c87	WO00127	1565ddad	9cc0842f				2026-10-04 06:01

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.225905+00:00", "is_mandatory": true, "process_step": "GRADE_SEPARATION", "schedule_code": "WO00129", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON",…	fdeb3cbe	WO00129	1565ddad	63f60409				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.226370+00:00", "is_mandatory": true, "process_step": "ACTIVATION", "schedule_code": "WO00130", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sche…	d27efea7	WO00130	1565ddad	7b6628b6				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.223079+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00123", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	e5078c1b	WO00123	1565ddad	cab75d16				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.222560+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00122", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	eb5cfa95	WO00122	1565ddad	caf7b3c2				2026-10-04 06:01

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.227293+00:00", "is_mandatory": true, "process_step": "QC", "schedule_code": "WO00132", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_sta…	46fa9719	WO00132	1565ddad	718a96c8				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:01	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:01:13.226897+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00131", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	b06db5bd	WO00131	1565ddad	ab8f805e				2026-10-04 06:01
COMPLETED	2026-10-04 06:01	{"source": "ac_routing_generate", "is_mandatory": true, "process_step": "ACTIVATION", "feed_recorded": true, "schedule_code": "EXEC-AC-121", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_status": "COMPLETED…	1c5f0dcf	EXEC-AC-121	1565ddad	64d837bc				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.894142+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00119", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	ae6fd6d4	WO00119	1acf8ae9	ab8f805e				2026-10-04 06:01

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.891920+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00114", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	9e46f148	WO00114	1acf8ae9	067fdcfe				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.892813+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00116", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	7c94486a	WO00116	1acf8ae9	0c7bf68a				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.890598+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00111", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	42b545d0	WO00111	1acf8ae9	cab75d16				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.891527+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00113", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	eaa8ce7c	WO00113	1acf8ae9	86967ce3				2026-10-04 06:01

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.893260+00:00", "is_mandatory": true, "process_step": "GRADE_SEPARATION", "schedule_code": "WO00117", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON",…	a3086b35	WO00117	1acf8ae9	63f60409				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.894582+00:00", "is_mandatory": true, "process_step": "QC", "schedule_code": "WO00120", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_sta…	f7606c6b	WO00120	1acf8ae9	718a96c8				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.893755+00:00", "is_mandatory": true, "process_step": "ACTIVATION", "schedule_code": "WO00118", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sche…	4d29db4a	WO00118	1acf8ae9	7b6628b6				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.891070+00:00", "is_mandatory": true, "process_step": "DESTONING", "schedule_code": "WO00112", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	77a18763	WO00112	1acf8ae9	a25971a3				2026-10-04 06:01

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.890041+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00110", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	52b325ca	WO00110	1acf8ae9	caf7b3c2				2026-10-04 06:01
NOT_STARTED	2026-10-04 06:00	{"source": "ac_routing_generate", "released_at": "2026-10-04T06:00:50.892310+00:00", "is_mandatory": true, "process_step": "DRYING", "schedule_code": "WO00115", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule…	4d7ef50f	WO00115	1acf8ae9	9cc0842f				2026-10-04 06:01
COMPLETED	2026-10-04 06:00	{"source": "ac_routing_generate", "is_mandatory": true, "process_step": "ACTIVATION", "feed_recorded": true, "schedule_code": "EXEC-AC-109", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule_status": "COMPLETED…	836ae3f9	EXEC-AC-109	1acf8ae9	64d837bc				2026-10-04 06:01
NOT_STARTED	2026-10-04 02:49	{"source": "ac_routing_generate", "released_at": "2026-10-04T02:50:03.198552+00:00", "is_mandatory": true, "process_step": "SCREENING", "schedule_code": "WO00107", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	4e178abe	WO00107	e58747dc	ab8f805e				2026-10-04 02:51

status	created_at	metadata	execution_id	execution_code	production_lot_id	routing_operation_id	equipment_id	started_at	completed_at	updated_at
NOT_STARTED	2026-10-04 02:49	{"source": "ac_routing_generate", "released_at": "2026-10-04T02:50:01.922342+00:00", "is_mandatory": true, "process_step": "KILN_FEED", "schedule_code": "WO00099", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sched…	851cd824	WO00099	e58747dc	cab75d16				2026-10-04 02:51
NOT_STARTED	2026-10-04 02:49	{"source": "ac_routing_generate", "released_at": "2026-10-04T02:50:02.723648+00:00", "is_mandatory": true, "process_step": "ACTIVATION", "schedule_code": "WO00104", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "sche…	217d39b5	WO00104	e58747dc	7b6628b6				2026-10-04 02:51
NOT_STARTED	2026-10-04 02:49	{"source": "ac_routing_generate", "released_at": "2026-10-04T02:50:02.405010+00:00", "is_mandatory": true, "process_step": "GRINDING", "schedule_code": "WO00102", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedu…	2ce8a607	WO00102	e58747dc	0c7bf68a				2026-10-04 02:51
NOT_STARTED	2026-10-04 02:49	{"source": "ac_routing_generate", "released_at": "2026-10-04T02:50:02.245929+00:00", "is_mandatory": true, "process_step": "DRYING", "schedule_code": "WO00101", "workspace_code": "GR_CARBONS_ACTIVATED_CARBON", "schedule…	410bfdec	WO00101	e58747dc	9cc0842f				2026-10-04 02:51

105 further row(s) omitted so the PDF stays readable.

exec.parameter_reading

Numeric / text readings captured on an execution.

Primary key: parameter_reading_id. Foreign keys: execution_id → exec.operation_execution(execution_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
parameter_reading_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
execution_id	UUID	NO	
parameter_code	VARCHAR(128)	NO	
parameter_name	VARCHAR(256)	YES	
reading_value	NUMERIC(18, 6)	YES	
reading_text	TEXT	YES	
uom	VARCHAR(16)	YES	
recorded_at	TIMESTAMP	NO	now()
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

No rows in scope (0 total in this filter).

qms.quality_check

QC header for a lot (lab, destoning bag QC, kiln output QC, dispatch QC).

Primary key: quality_check_id. Foreign keys: production_lot_id → trn.production_lot(production_lot_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
quality_check_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
check_code	VARCHAR(64)	NO	
production_lot_id	UUID	NO	
check_type	VARCHAR(64)	NO	'IN_PROCESS'::character varying
status	VARCHAR(32)	NO	'OPEN'::character varying
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (1 of 1 row(s) in GR_CARBONS / catalog scope):

check_code	status	created_at	metadata	quality_check_id	production_lot_id	check_type	updated_at	workspace_id
DISP-AC-1	Shipped	2026-10-05 05:53	{"lines": [{"lot_id": "LOT-AC-12", "part_id": "FG-AC-1", "accepted_quantity": "1000", "dispatch_quantity": "1000", "rejected_quantity": "0"}], "remarks": null, "order_id": "CUST_PO_5", "grade_code": "FG-AC-1", "customer…	0627cddd	2b665247	DISPATCH	2026-10-05 05:53	e7000000

qms.quality_result

QC characteristics; bag lists and destinations often sit in metadata JSON.

Primary key: quality_result_id. Foreign keys: quality_check_id → qms.quality_check(quality_check_id); tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
quality_result_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
quality_check_id	UUID	NO	
characteristic_code	VARCHAR(128)	NO	
characteristic_name	VARCHAR(256)	YES	
measured_value	NUMERIC(18, 6)	YES	
measured_text	TEXT	YES	
uom	VARCHAR(16)	YES	
result_status	VARCHAR(32)	NO	'PASS'::character varying
recorded_at	TIMESTAMP	NO	now()
metadata	JSONB	NO	'{}'::jsonb
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

No rows in scope (0 total in this filter).

kg.triple_store_staging

Triplet ontology: subject / predicate / object for inward and lot URIs.

Primary key: triple_id. Foreign keys: tenant_id → core.tenant(tenant_id); workspace_id → core.project_workspace(workspace_id).

Column	Type	Null	Default
triple_id	UUID	NO	gen_random_uuid()
tenant_id	UUID	NO	
subject_uri	TEXT	NO	
predicate_uri	TEXT	NO	
object_uri	TEXT	YES	
object_literal	TEXT	YES	
object_datatype	VARCHAR(128)	YES	
source_table	VARCHAR(128)	YES	

Column	Type	Null	Default
source_id	UUID	YES	
export_status	VARCHAR(32)	NO	'PENDING'::character varying
created_at	TIMESTAMP	NO	now()
updated_at	TIMESTAMP	NO	now()
workspace_id	UUID	YES	

Data (60 of 1163 row(s) in GR_CARBONS / catalog scope):

subject_uri	predicate_uri	object_uri	object_literal	source_table	created_at	triple_id	object_datatype	source_id	export_status	updated_at
urn:nyra:ac:inward:ORD-AC-22	urn:nyra:predicate:captures	urn:nyra:ac:parameter:OUR_WEIGHT_KG	2300	trn.customer_order	2026-10-05 06:58	cdd529f0	http://www.w3.org/2001/XMLSchema#decimal	30070574	PENDING	2026-10-05 06:58
urn:nyra:ac:inward:ORD-AC-22	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SUPPLIER_WEIGHT_KG	2300	trn.customer_order	2026-10-05 06:58	6e7fe800	http://www.w3.org/2001/XMLSchema#decimal	30070574	PENDING	2026-10-05 06:58
urn:nyra:ac:inward:ORD-AC-22	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DC_NUMBER	DC5678	trn.customer_order	2026-10-05 06:58	1a8798e0	http://www.w3.org/2001/XMLSchema#string	30070574	PENDING	2026-10-05 06:58
urn:nyra:ac:inward:ORD-AC-22	urn:nyra:predicate:captures	urn:nyra:ac:parameter:RECEIVED_QTY_KG	2300	trn.customer_order	2026-10-05 06:57	3798708d	http://www.w3.org/2001/XMLSchema#decimal	30070574	PENDING	2026-10-05 06:57
urn:nyra:ac:supplier:SUP-AC-1	urn:nyra:predicate:supplies	urn:nyra:ac:inward:ORD-AC-22		trn.customer_order	2026-10-05 06:57	8dcc815c		30070574	PENDING	2026-10-05 06:57
urn:nyra:ac:screenBag:SC_051026_00006	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_BAG_REF	SC_051026_00006	cfg.resource	2026-10-05 05:03	c9fe564e	http://www.w3.org/2001/XMLSchema#string	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_GRADE	+30	cfg.resource	2026-10-05 05:03	4aecb910	http://www.w3.org/2001/XMLSchema#decimal	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_OUTPUT_KG	333	cfg.resource	2026-10-05 05:03	501ffe37	http://www.w3.org/2001/XMLSchema#decimal	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00006	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_OUTPUT_KG	333	cfg.resource	2026-10-05 05:03	c40da428	http://www.w3.org/2001/XMLSchema#decimal	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_INPUT_BAG_REF	DS_051026_00005	cfg.resource	2026-10-05 05:03	50e9157b	http://www.w3.org/2001/XMLSchema#string	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_BAG_REF	SC_051026_00006	cfg.resource	2026-10-05 05:03	2a538b3c	http://www.w3.org/2001/XMLSchema#string	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00006	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_GRADE	+30	cfg.resource	2026-10-05 05:03	796efa4c	http://www.w3.org/2001/XMLSchema#decimal	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00006	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_INPUT_BAG_REF	DS_051026_00005	cfg.resource	2026-10-05 05:03	956b0a80	http://www.w3.org/2001/XMLSchema#string	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00006	urn:nyra:predicate:generatedFrom	urn:nyra:ac:destonBag:DS_051026_00005		cfg.resource	2026-10-05 05:03	be239426		57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00006	urn:nyra:predicate:contained_in	urn:nyra:ac:postActivation:RES-POSTACT-01:screening		cfg.resource	2026-10-05 05:03	d7d421d0		57261196	PENDING	2026-10-05 05:03

subject_uri	predicate_uri	object_uri	object_literal	source_table	created_at	triple_id	object_datatype	source_id	export_status	updated_at
urn:nyra:ac:screenBag:SC_051026_00005	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_INPUT_BAG_REF	DS_051026_00005	cfg.resource	2026-10-05 05:03	d711efa5	http://www.w3.org/2001/XMLSchema#string	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_OUTPUT_KG	188	cfg.resource	2026-10-05 05:03	9bac0093	http://www.w3.org/2001/XMLSchema#decimal	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_BAG_REF	SC_051026_00005	cfg.resource	2026-10-05 05:03	84a6dbf6	http://www.w3.org/2001/XMLSchema#string	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_GRADE	+20	cfg.resource	2026-10-05 05:03	ad7d789a	http://www.w3.org/2001/XMLSchema#decimal	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00005	urn:nyra:predicate:contained_in	urn:nyra:ac:postActivation:RES-POSTACT-01:screening		cfg.resource	2026-10-05 05:03	f7f97611		57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00005	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_BAG_REF	SC_051026_00005	cfg.resource	2026-10-05 05:03	65406a53	http://www.w3.org/2001/XMLSchema#string	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_INPUT_BAG_REF	DS_051026_00005	cfg.resource	2026-10-05 05:03	04495e57	http://www.w3.org/2001/XMLSchema#string	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00005	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_OUTPUT_KG	188	cfg.resource	2026-10-05 05:03	bc446327	http://www.w3.org/2001/XMLSchema#decimal	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00005	urn:nyra:predicate:generatedFrom	urn:nyra:ac:destonBag:DS_051026_00005		cfg.resource	2026-10-05 05:03	45a66e43		57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:screenBag:SC_051026_00005	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SCREENING_GRADE	+20	cfg.resource	2026-10-05 05:03	c33dbfc9	http://www.w3.org/2001/XMLSchema#decimal	57261196	PENDING	2026-10-05 05:03
urn:nyra:ac:inward:ORD-AC-21	urn:nyra:predicate:captures	urn:nyra:ac:parameter:OUR_WEIGHT_KG	2500	trn.customer_order	2026-10-05 02:55	017fbc02	http://www.w3.org/2001/XMLSchema#decimal	d06819bc	PENDING	2026-10-05 02:55
urn:nyra:ac:inward:ORD-AC-21	urn:nyra:predicate:captures	urn:nyra:ac:parameter:SUPPLIER_WEIGHT_KG	2500	trn.customer_order	2026-10-05 02:55	4fcb15ef	http://www.w3.org/2001/XMLSchema#decimal	d06819bc	PENDING	2026-10-05 02:55
urn:nyra:ac:inward:ORD-AC-21	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DC_NUMBER	DC12333	trn.customer_order	2026-10-05 02:55	b63697cb	http://www.w3.org/2001/XMLSchema#string	d06819bc	PENDING	2026-10-05 02:55
urn:nyra:ac:inward:ORD-AC-21	urn:nyra:predicate:captures	urn:nyra:ac:parameter:RECEIVED_QTY_KG	2500	trn.customer_order	2026-10-05 02:55	85d03118	http://www.w3.org/2001/XMLSchema#decimal	d06819bc	PENDING	2026-10-05 02:55
urn:nyra:ac:supplier:SUP-AC-1	urn:nyra:predicate:supplies	urn:nyra:ac:inward:ORD-AC-21		trn.customer_order	2026-10-05 02:55	38f91753		d06819bc	PENDING	2026-10-05 02:55
urn:nyra:ac:destonBag:DS_041026_00010	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTON_QC_DESTINATION	Screening	cfg.resource	2026-10-04 20:56	d0067373	http://www.w3.org/2001/XMLSchema#string	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00009	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTON_QC_DESTINATION	De-Dusting	cfg.resource	2026-10-04 20:56	5f63781f	http://www.w3.org/2001/XMLSchema#string	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTONING_BAG_REF	DS_041026_00010	cfg.resource	2026-10-04 20:56	00c92414	http://www.w3.org/2001/XMLSchema#string	3c018e7d	PENDING	2026-10-04 20:56

subject_uri	predicate_uri	object_uri	object_literal	source_table	created_at	triple_id	object_datatype	source_id	export_status	updated_at
urn:nyra:ac:destonBag:DS_041026_00010	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTON_OUTPUT_KG	55555	cfg.resource	2026-10-04 20:56	0638934a	http://www.w3.org/2001/XMLSchema#decimal	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00010	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTON_INPUT_BAG_REF		cfg.resource	2026-10-04 20:56	1e7661af	http://www.w3.org/2001/XMLSchema#string	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00010	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTONING_BAG_REF	DS_041026_00010	cfg.resource	2026-10-04 20:56	59d168ca	http://www.w3.org/2001/XMLSchema#string	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00010	urn:nyra:predicate:contained_in	urn:nyra:ac:destoner:RES-DESTON-01		cfg.resource	2026-10-04 20:56	ae692a78		3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTON_OUTPUT_KG	55555	cfg.resource	2026-10-04 20:56	1c18de82	http://www.w3.org/2001/XMLSchema#decimal	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00009	urn:nyra:predicate:generatedFrom	urn:nyra:ac:kilnOutBag:KB_051026_00008		cfg.resource	2026-10-04 20:56	b55812a4		3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00009	urn:nyra:predicate:generatedFrom	urn:nyra:ac:kilnOutBag:KB_041026_00002		cfg.resource	2026-10-04 20:56	d5d35560		3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTONING_BAG_REF	DS_041026_00009	cfg.resource	2026-10-04 20:56	ddc008a0	http://www.w3.org/2001/XMLSchema#string	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00009	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTON_OUTPUT_KG	1000	cfg.resource	2026-10-04 20:56	85baf30b	http://www.w3.org/2001/XMLSchema#decimal	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00009	urn:nyra:predicate:generatedFrom	urn:nyra:ac:kilnOutBag:KB_041026_00003		cfg.resource	2026-10-04 20:56	7ae5cf8d		3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00009	urn:nyra:predicate:contained_in	urn:nyra:ac:destoner:RES-DESTON-01		cfg.resource	2026-10-04 20:56	917d433f		3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00009	urn:nyra:predicate:generatedFrom	urn:nyra:ac:kilnOutBag:KB_051026_00009		cfg.resource	2026-10-04 20:56	2b848f87		3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:lot:LOT-AC-23	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTON_OUTPUT_KG	1000	cfg.resource	2026-10-04 20:56	8c1e4366	http://www.w3.org/2001/XMLSchema#decimal	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00009	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTONING_BAG_REF	DS_041026_00009	cfg.resource	2026-10-04 20:56	7a744bd9	http://www.w3.org/2001/XMLSchema#string	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:destonBag:DS_041026_00009	urn:nyra:predicate:captures	urn:nyra:ac:parameter:DESTON_INPUT_BAG_REF	KB_051026_00008,KB_051026_00009,KB_041026_00002,KB_041026_00003	cfg.resource	2026-10-04 20:56	402d13f0	http://www.w3.org/2001/XMLSchema#string	3c018e7d	PENDING	2026-10-04 20:56
urn:nyra:ac:kilnOutBag:KB_041026_00011	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KILN_OUT_4_8_KG	4	cfg.resource	2026-10-04 20:55	90328cd3	http://www.w3.org/2001/XMLSchema#decimal	0d4428e7	PENDING	2026-10-04 20:55
urn:nyra:ac:kilnOutBag:KB_041026_00011	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KILN_OUT_3_4_KG	4	cfg.resource	2026-10-04 20:55	f98e9ff4	http://www.w3.org/2001/XMLSchema#decimal	0d4428e7	PENDING	2026-10-04 20:55
urn:nyra:ac:kilnOutBag:KB_041026_00011	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KILN_OUT_PLUS_3_KG	4	cfg.resource	2026-10-04 20:55	85d7d508	http://www.w3.org/2001/XMLSchema#decimal	0d4428e7	PENDING	2026-10-04 20:55
urn:nyra:ac:kilnOutBag:KB_041026_00012	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KILN_OUT_8_12_KG	4	cfg.resource	2026-10-04 20:55	3f232492	http://www.w3.org/2001/XMLSchema#decimal	0d4428e7	PENDING	2026-10-04 20:55

subject_uri	predicate_uri	object_uri	object_literal	source_table	created_at	triple_id	object_datatype	source_id	export_status	updated_at
urn:nyra:ac:kilnOutBag:KB_041026_00012	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KILN_OUT_4_8_KG	4	cfg.resource	2026-10-04 20:55	1b5e692c	http://www.w3.org/2001/XMLSchema#decimal	0d4428e7	PENDING	2026-10-04 20:55
urn:nyra:ac:kilnOutBag:KB_041026_00012	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KILN_OUT_3_4_KG	4	cfg.resource	2026-10-04 20:55	c2bc1e0d	http://www.w3.org/2001/XMLSchema#decimal	0d4428e7	PENDING	2026-10-04 20:55
urn:nyra:ac:kilnOutBag:KB_041026_00012	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KILN_OUT_PLUS_3_KG	4	cfg.resource	2026-10-04 20:55	9123f499	http://www.w3.org/2001/XMLSchema#decimal	0d4428e7	PENDING	2026-10-04 20:55
urn:nyra:ac:inwardBag:RO_BAG_014	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KFQC_10_12_KG	4	trn.customer_order	2026-10-04 20:55	14507111	http://www.w3.org/2001/XMLSchema#decimal	63d91e23	PENDING	2026-10-04 20:55
urn:nyra:ac:inwardBag:RO_BAG_014	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KFQC_6_8_KG	4	trn.customer_order	2026-10-04 20:55	e6b96e42	http://www.w3.org/2001/XMLSchema#decimal	63d91e23	PENDING	2026-10-04 20:55
urn:nyra:ac:inwardBag:RO_BAG_014	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KFQC_3_6_KG	4	trn.customer_order	2026-10-04 20:55	71c91a33	http://www.w3.org/2001/XMLSchema#decimal	63d91e23	PENDING	2026-10-04 20:55
urn:nyra:ac:inwardBag:RO_BAG_014	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KFQC_2_3_KG	4	trn.customer_order	2026-10-04 20:55	7b482ec9	http://www.w3.org/2001/XMLSchema#decimal	63d91e23	PENDING	2026-10-04 20:55
urn:nyra:ac:inwardBag:RO_BAG_014	urn:nyra:predicate:captures	urn:nyra:ac:parameter:KFQC_PLUS_2_KG	4	trn.customer_order	2026-10-04 20:55	2aafece2	http://www.w3.org/2001/XMLSchema#decimal	63d91e23	PENDING	2026-10-04 20:55

1103 further row(s) omitted so the PDF stays readable.

4. Triplet predicates in use (GR_CARBONS)

These are the ontology facts already written. Inward uses urn:nyra:ac:inward:{order}; lots use urn:nyra:ac:lot:{lot}. Do not replace this store with a parallel capture model.

Predicate	Count
urn:nyra:predicate:captures	909
urn:nyra:predicate:containedIn	112
urn:nyra:predicate:generatedFrom	45
urn:nyra:predicate:contained_in	33
urn:nyra:predicate:chargedTo	27
urn:nyra:predicate:supplies	19
urn:nyra:predicate:assignedTo	18